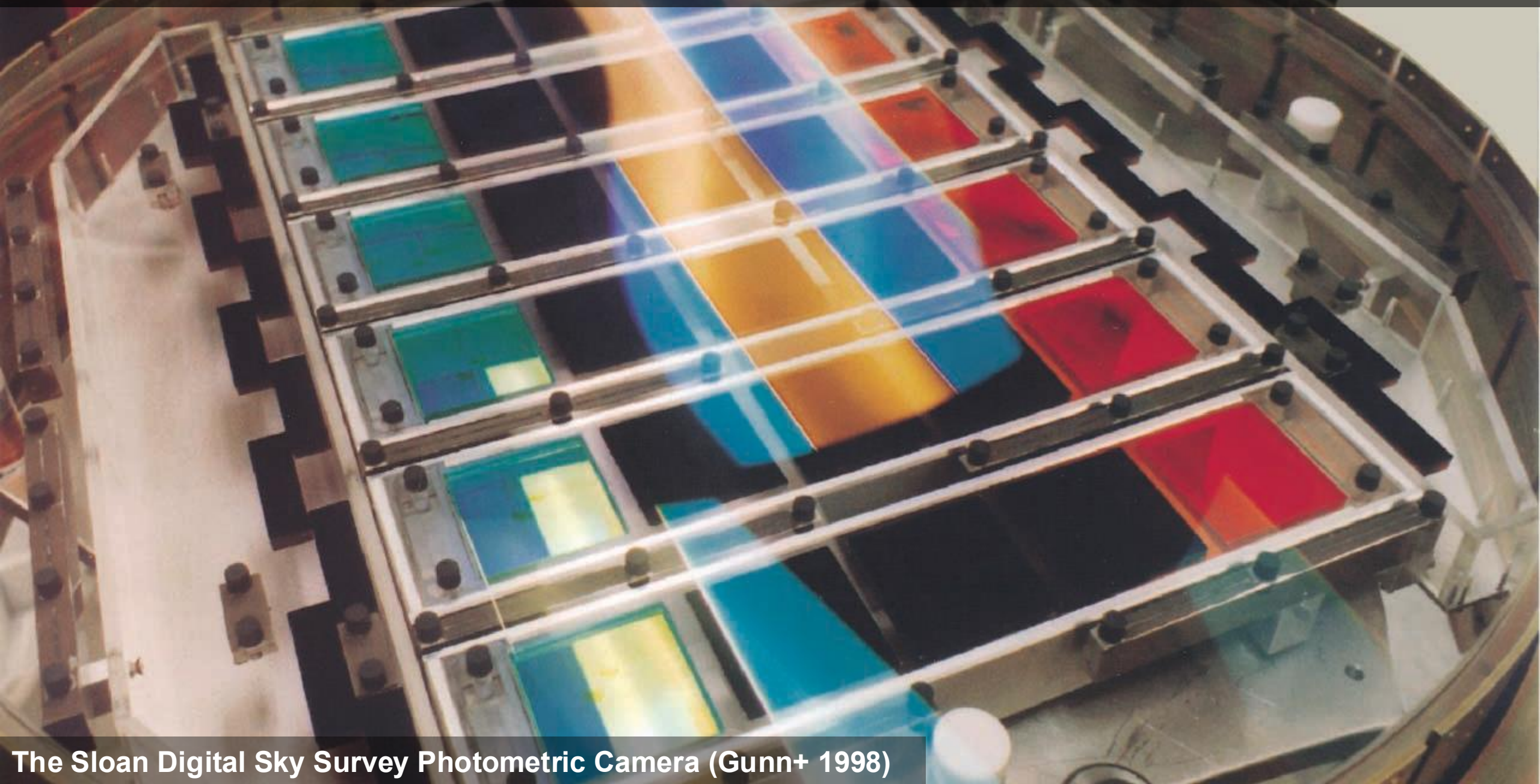


ASTR 302 Fall 2026: Detectors



The Sloan Digital Sky Survey Photometric Camera (Gunn+ 1998)

Agenda – September 14, 2026

- Class Logistics
- Introducing Detectors and Charge Coupled Devices (CCDs)

Recommended Reading: Birney Chapter 8
Other Resources: "Detection of Light" - George Rieke

Class Logistics

- 5 – minute presentations + questions describing your project **(9/21) (5% of your Final Project Grade)**
- **Each Team Pick a New NAME ASAP**
- **Observing Schedule and Timing**
 - 61 inch teams (2:30pm)
 - 90 inch teams (2:00pm)
 - **Pick up Location in FRONT OF STEWARD**
- Will send email about observing logistics later.
- Lab due Friday (9/18 at 11:59pm) (Neptune and Uranus)

Detecting and Recording Light



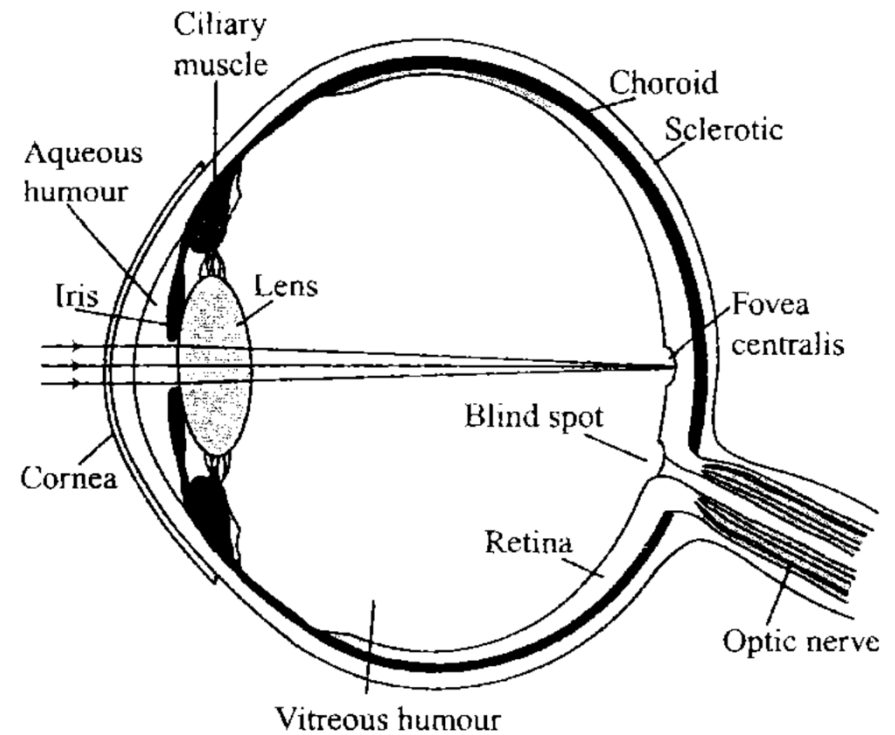
An astronomical detector receives photons from a source and produces a signal. It is one of the most important parts of an instrument.

Detecting Light: Your Eye

The original detector is the human eye. Our eyes have a lens that reimages our surroundings.

We can observe celestial objects behind a telescope with an eye piece.

The eye can adjust to varying brightness levels, but its not objective.



Detecting and Recording Light: Photography

Photographic plates were glass plates coated with emulsions to be light sensitive.

Photographic plates were extremely inefficient at capturing all the light incident on the plate.

The physical size of photographic plates were quite large compared to detectors we have now.



Detecting and Recording Light: CCDs

Charge-Coupled Devices (CCD) use the photoelectric effect to convert photons into electronic signals.

It is a 2-D array of **pixels**. Each pixel can convert photons into electric charge and store them.

Electric charge is transferred across the array to output registers near the edge of the detector.



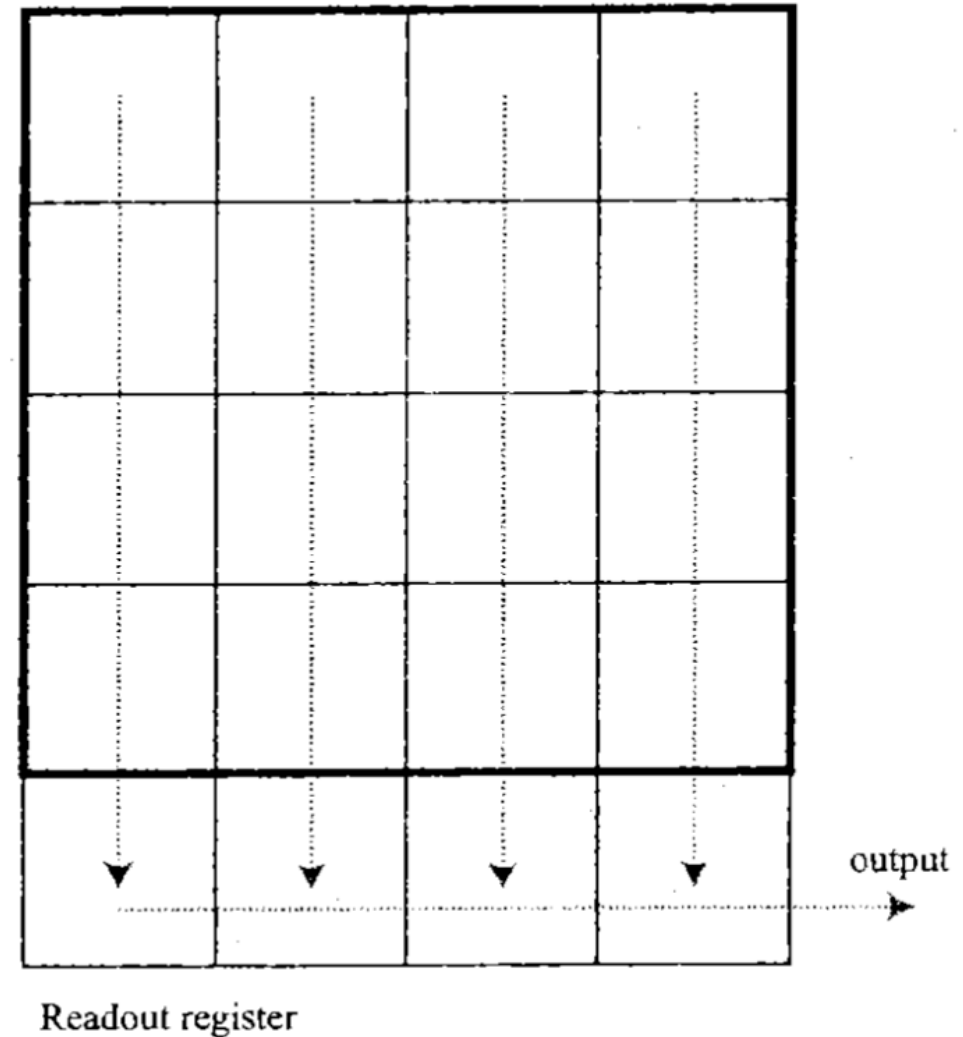
Kepler used 42 CCDs

Detecting and Recording Light: CCD Readout

Electric charge is transferred across the array to output registers near the edge of the detector.

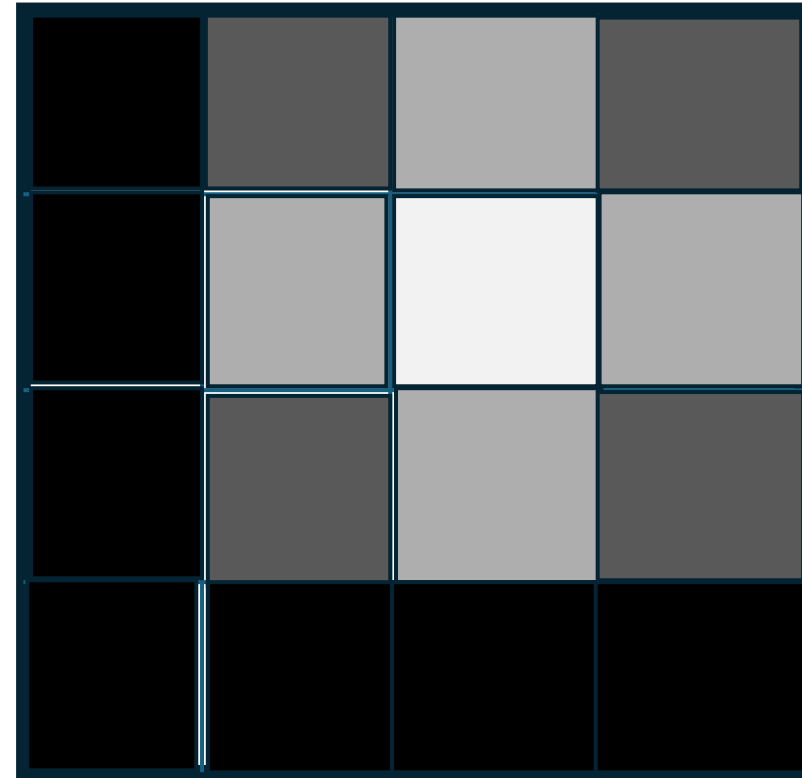
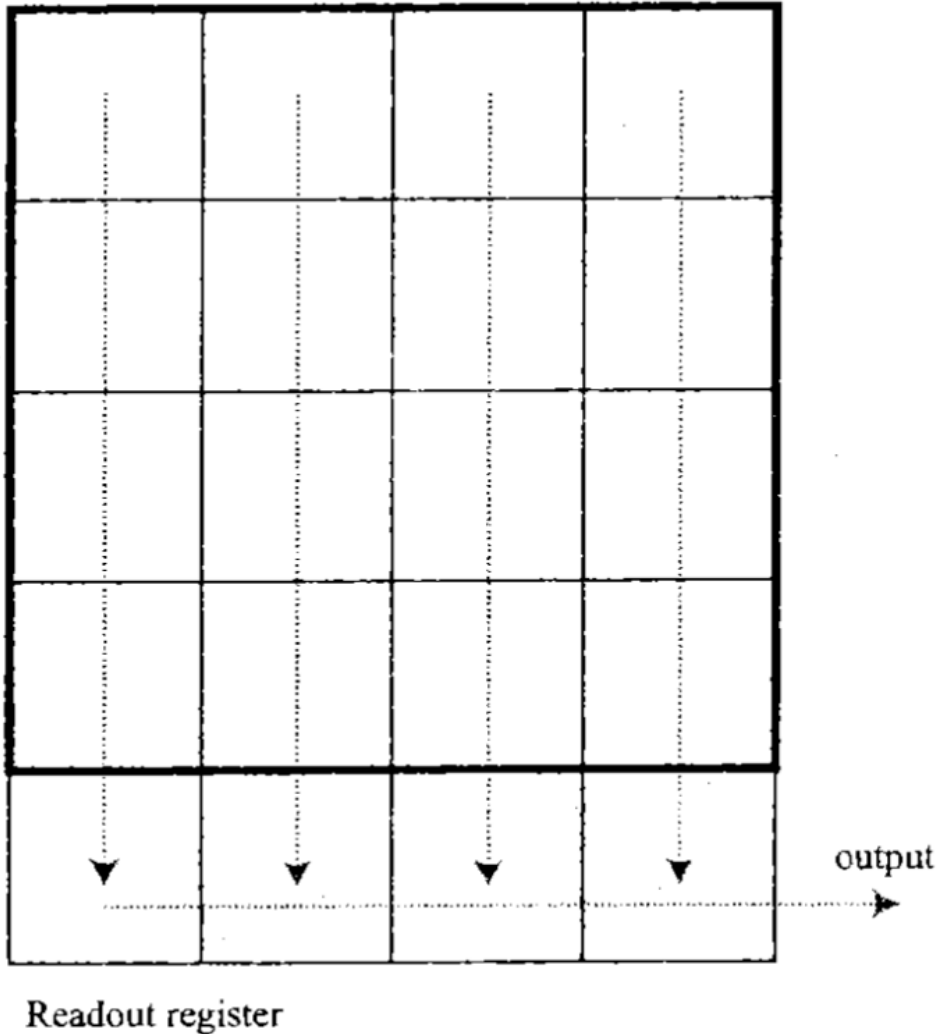
This way of reading out is kind of slow if you have a relatively large CCD.

There is also a maximum amount of charge that can remain on each pixel. This can be tuned a little.



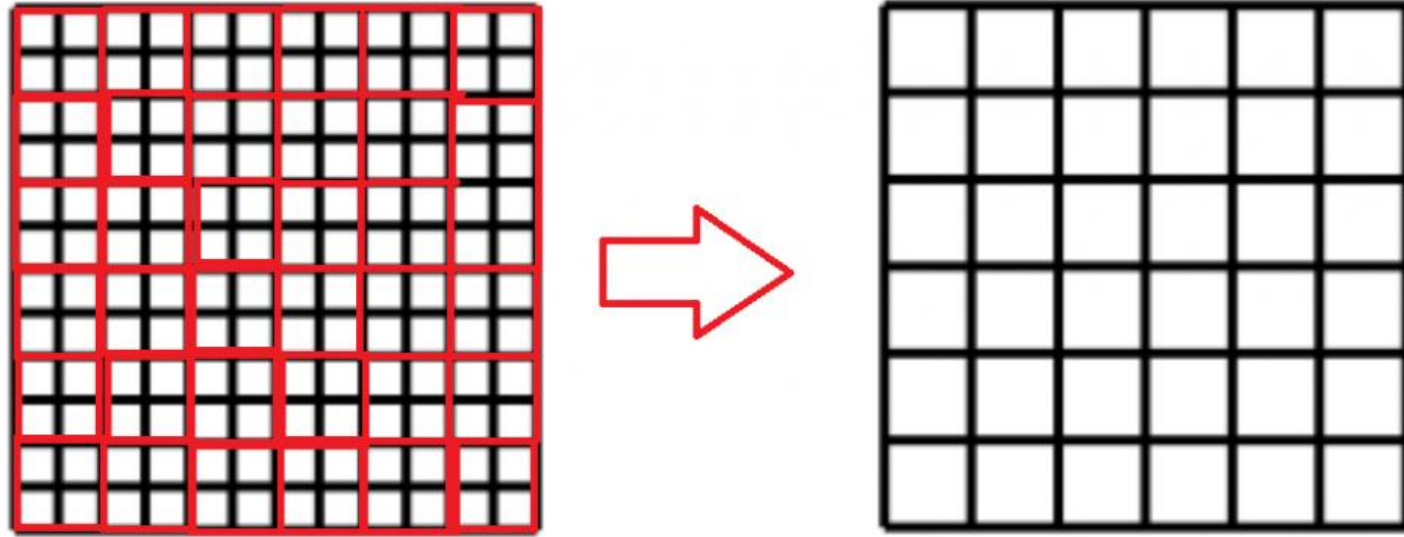
Detecting and Recording Light: CCD Readout

DS9 / Python View



The images you manipulate on your computer are related to the pixels you read out on the detector/CCD.

Detecting and Recording Light: CCD Readout

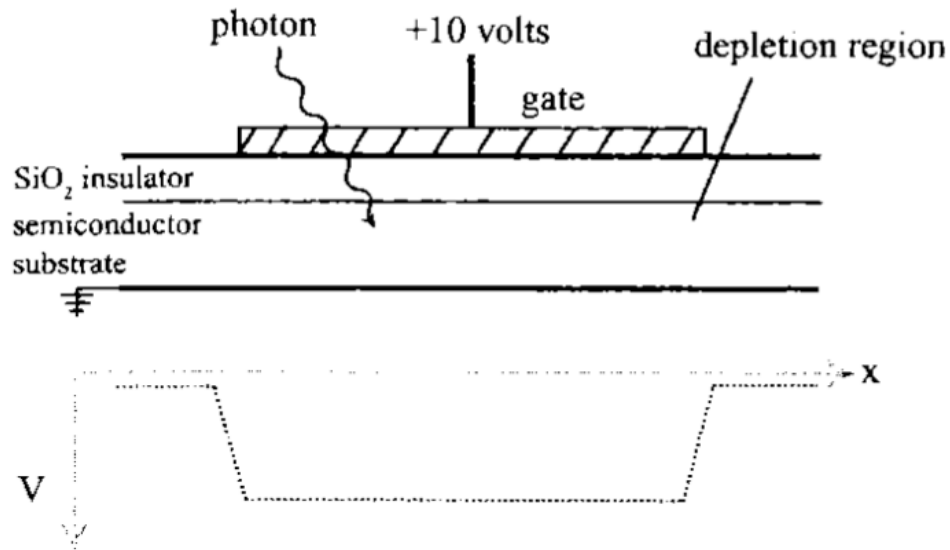


2 X 2 Binning

You can combine the information from several nearby pixels by “binning”. The Mont4K detector can use binning.

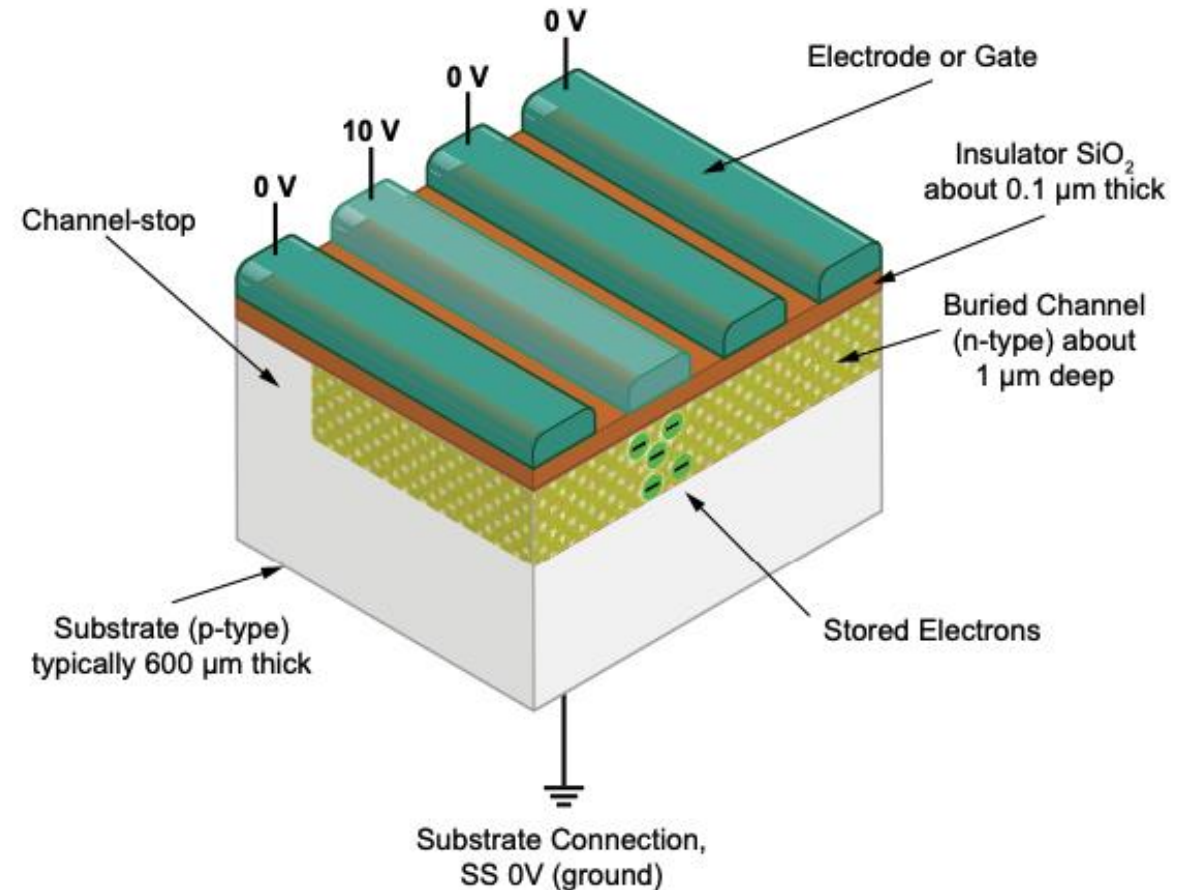
Detecting and Recording Light: CCD pixels

Single CCD Pixel (Book)



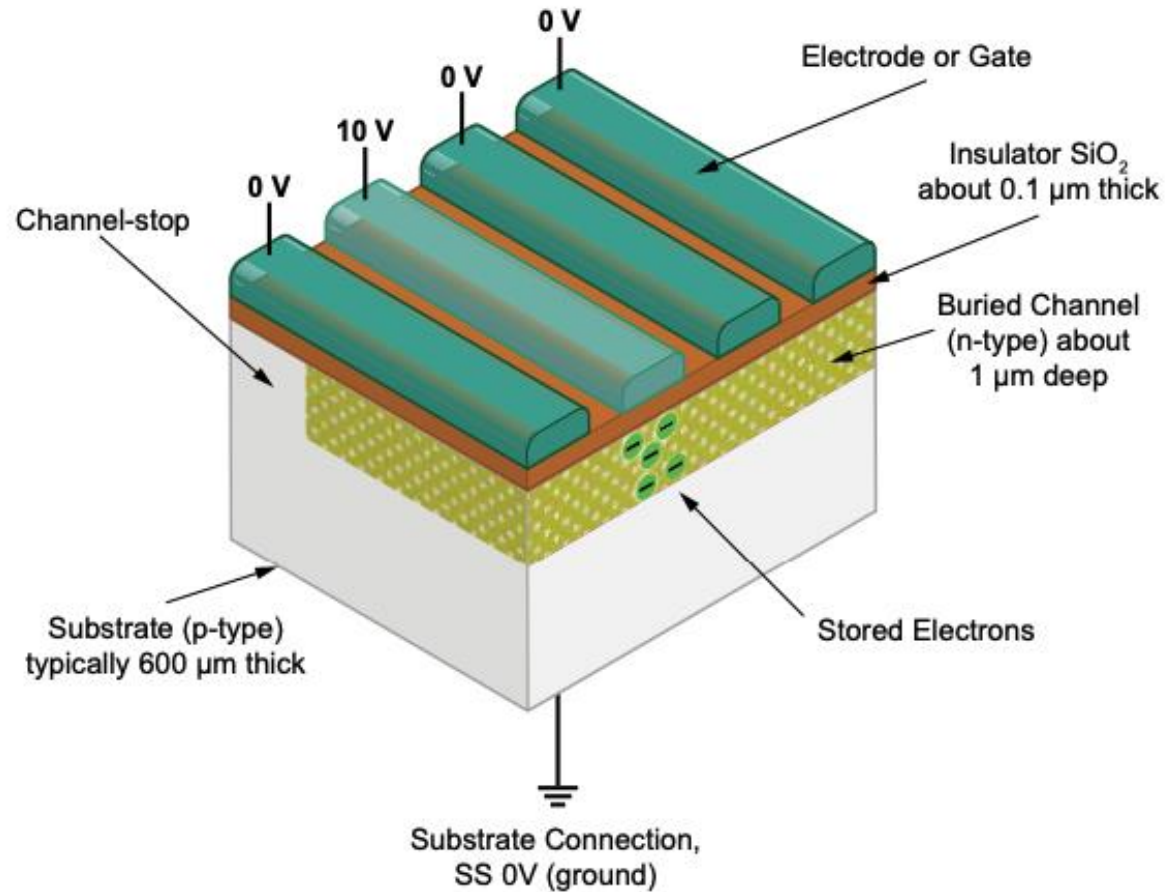
Front illuminated pixel

Single CCD Pixel (Teledyne diagram, 4 Gates)

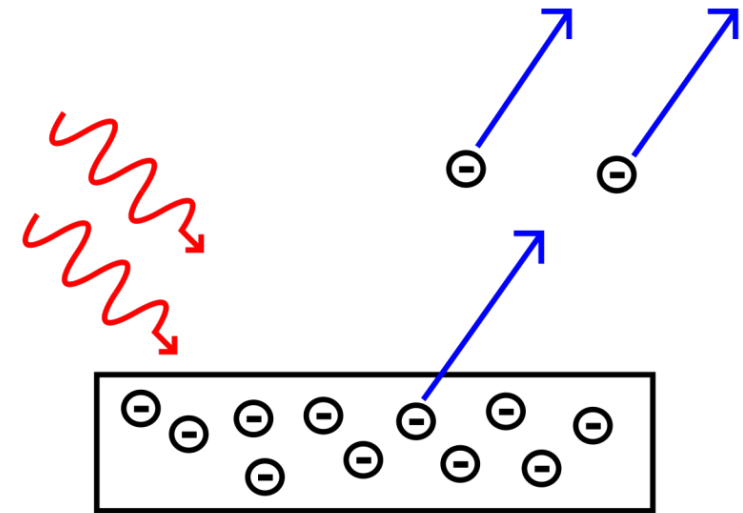


Detecting and Recording Light: CCD pixels

Single CCD Pixel

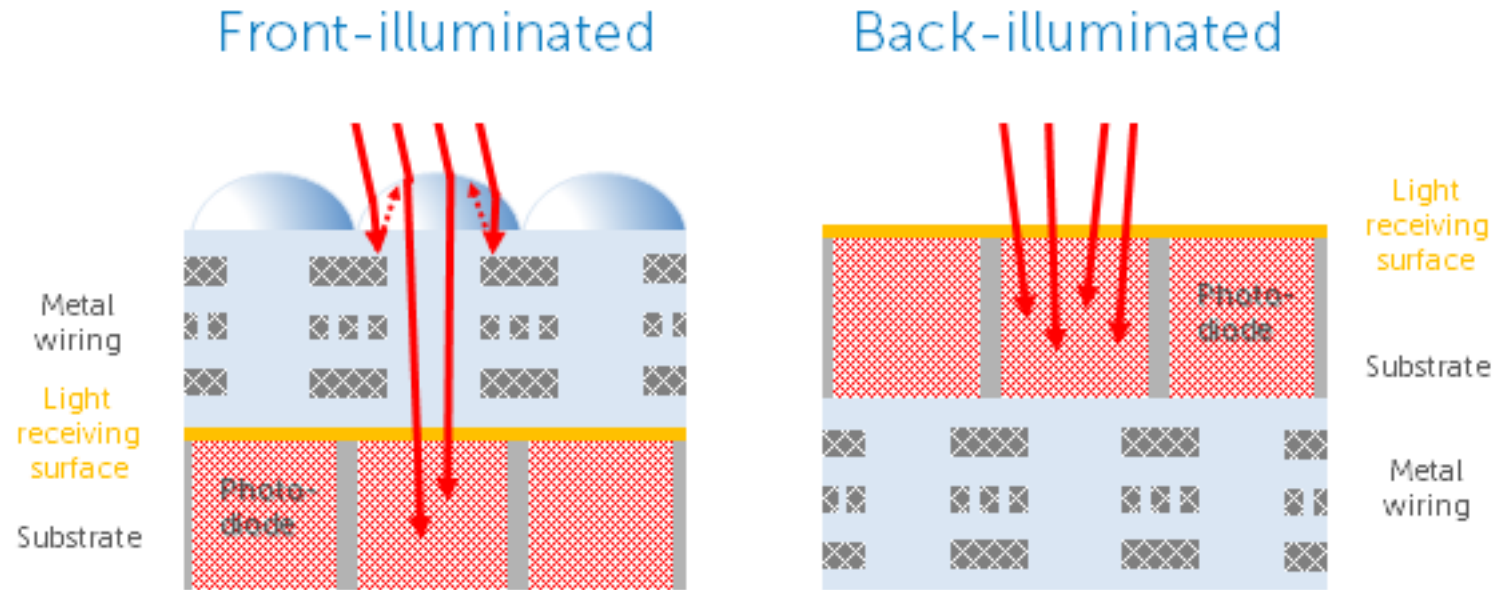


Photoelectric Effect



- Because of the photoelectric effect, photons hitting the substrate releases electrons.

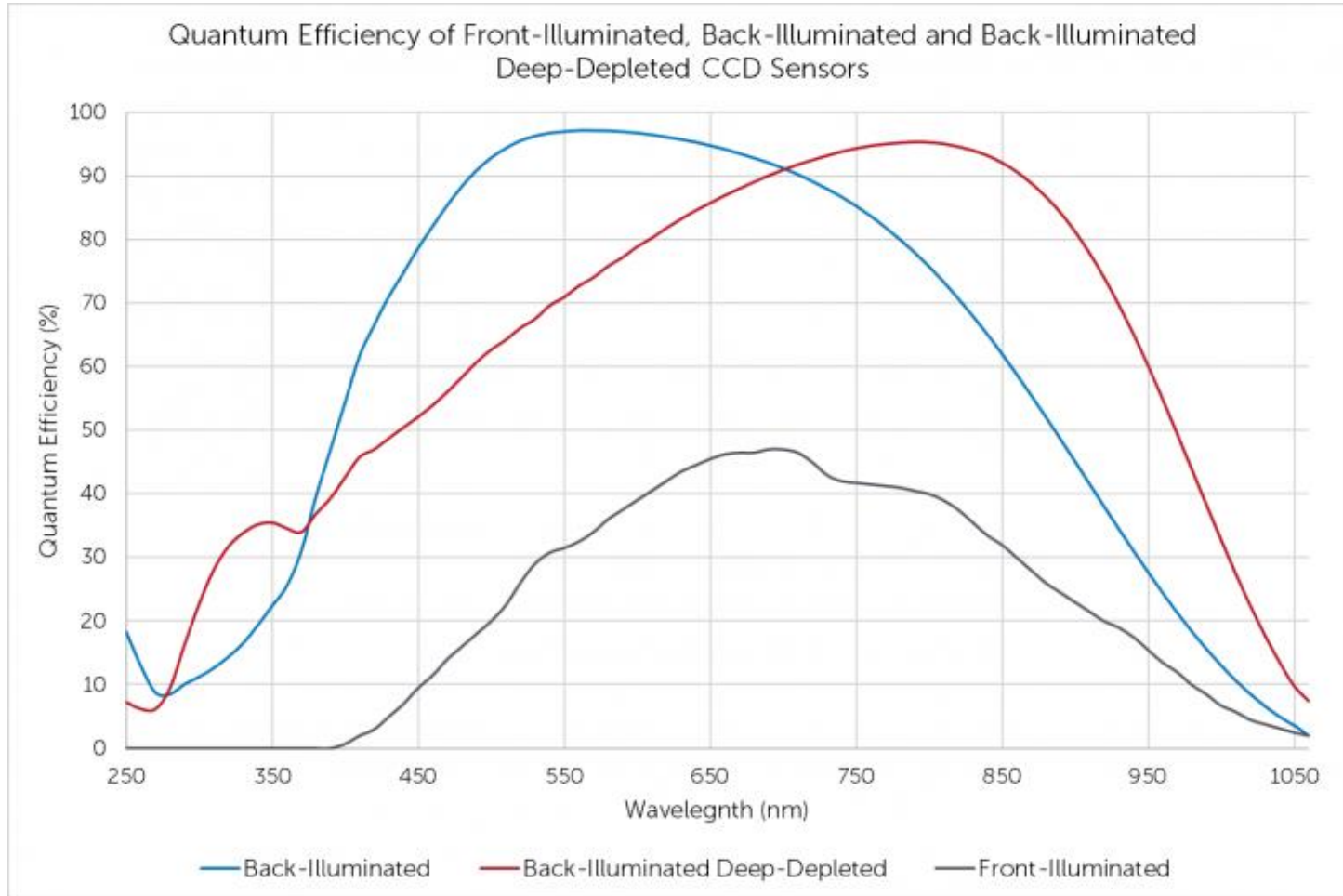
Detecting and Recording Light: CCD pixels



Front Illuminated pixels/CCDs have gates on top of the substrate. This limits light sensitivity in some wavelengths.

Back Illuminated pixels/CCDs have gates underneath the substrate and are more efficient at capturing light. (**SDSS detectors**)

Quantum Efficiency of Detectors

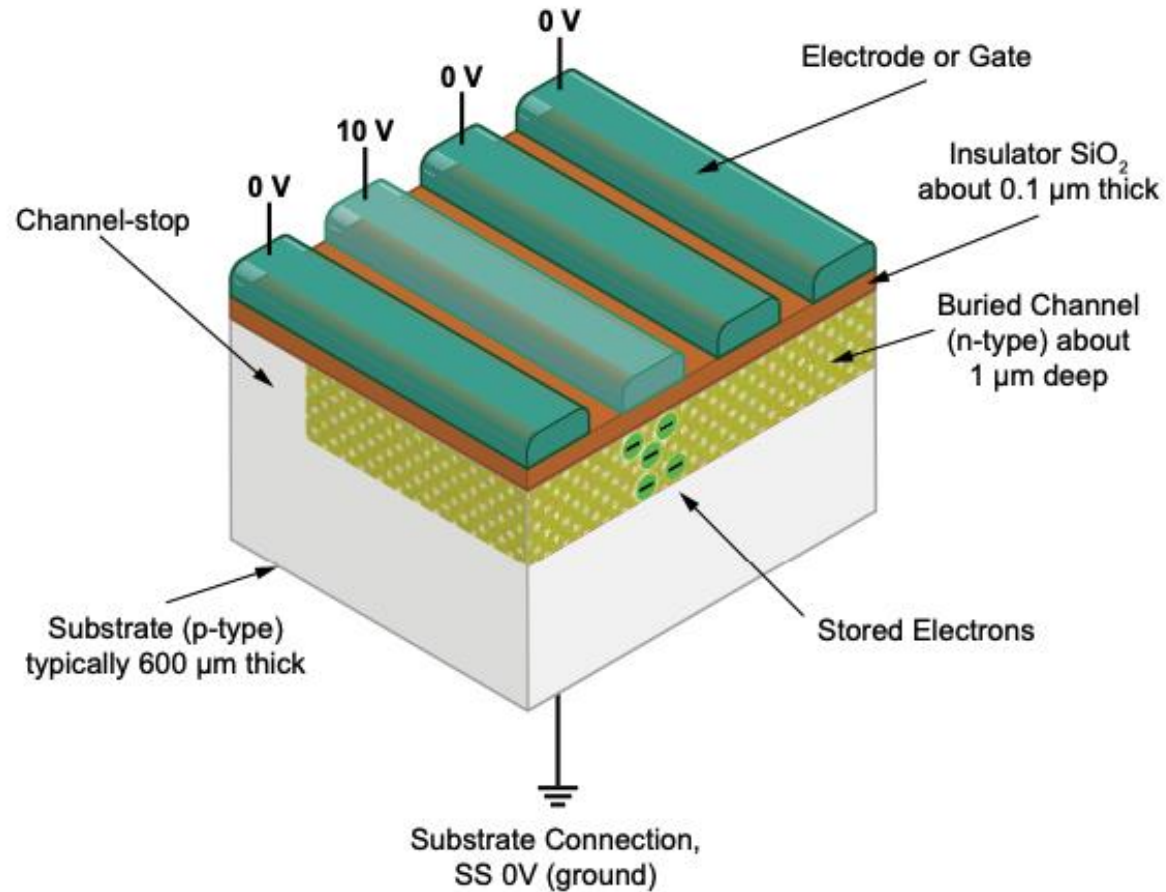


Quantum efficiency (QE) is the percentage of photons a detector can convert to electrons.

The pixel structure, thickness, and materials used impacts a detector's quantum efficiency.

Detecting and Recording Light: CCD pixels

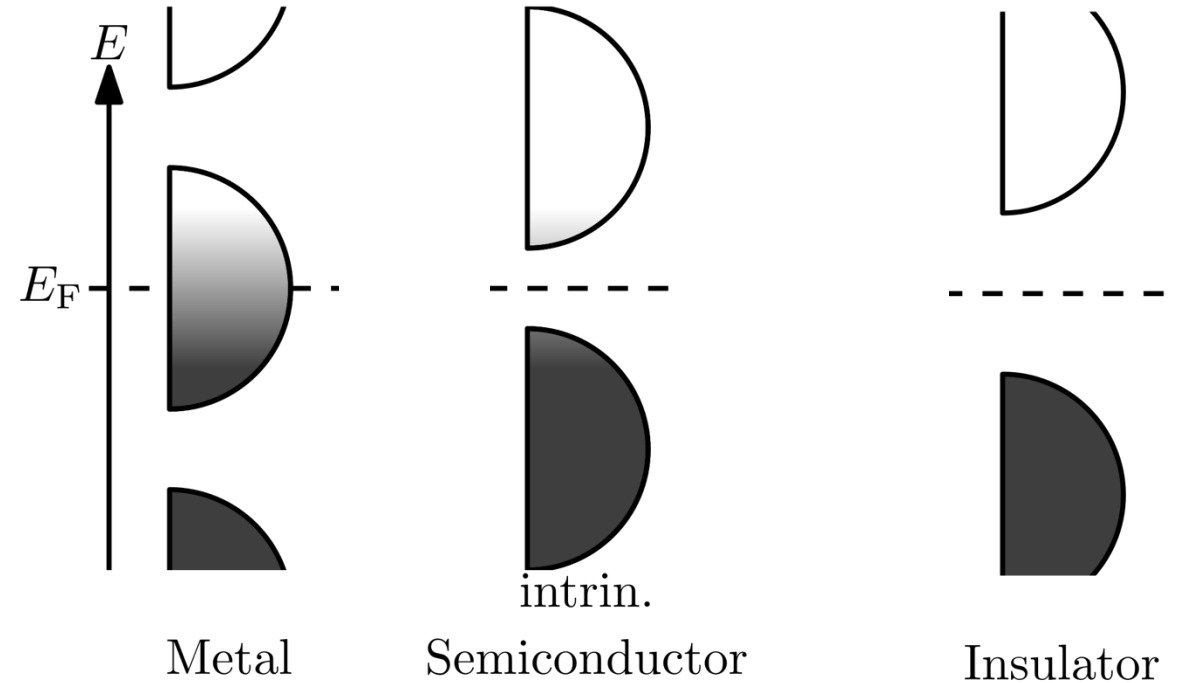
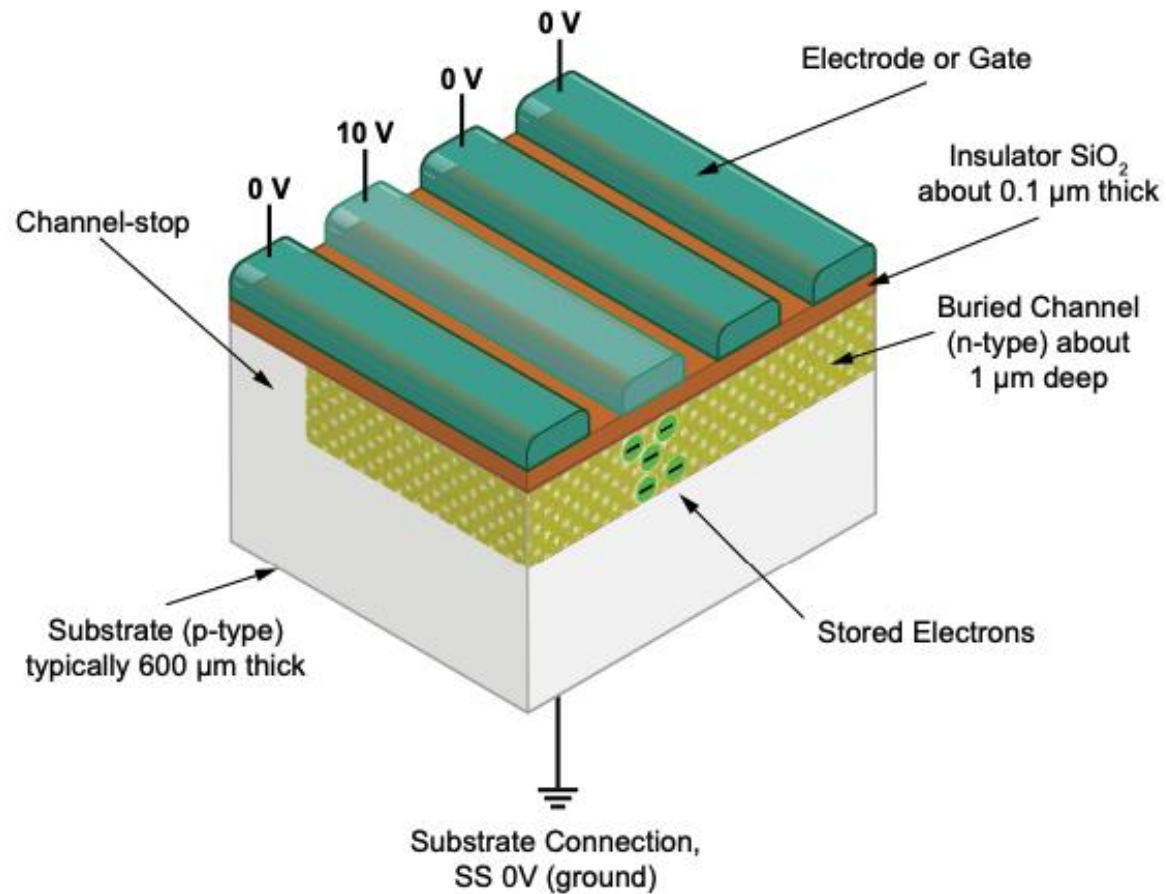
Single CCD Pixel



- Doped semiconductor layers (p-type and n-type) work as a photodiode where a photon can be measured based on the amount of electrons present.

Detecting and Recording Light: CCD pixels

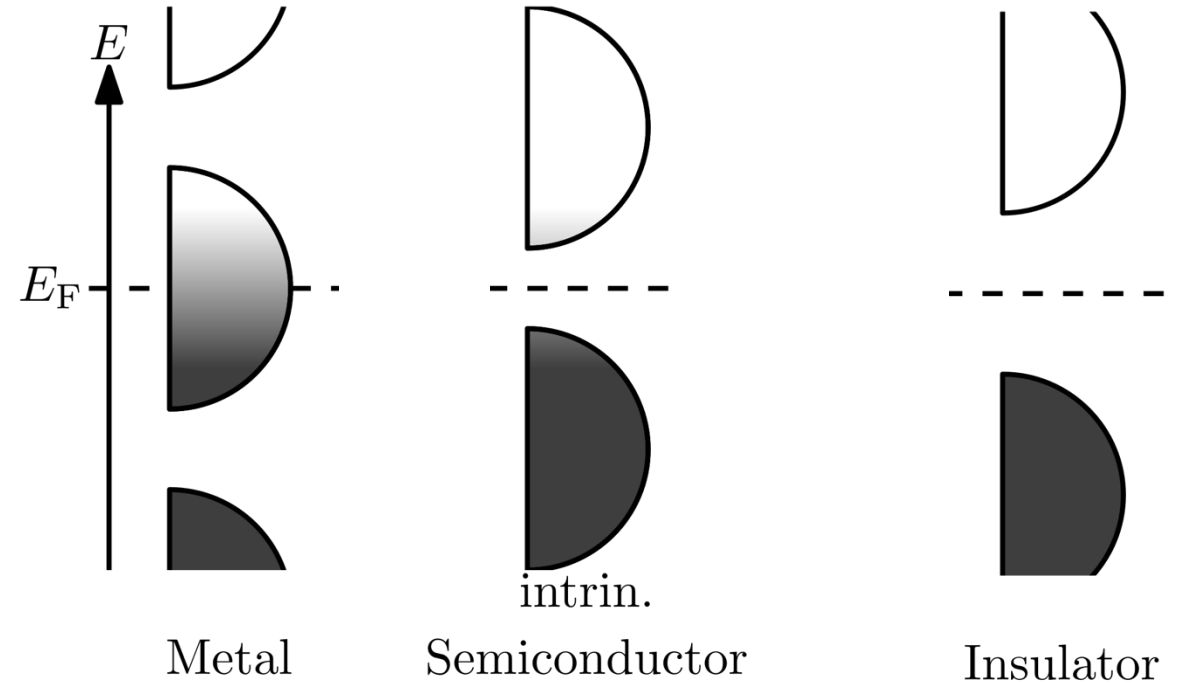
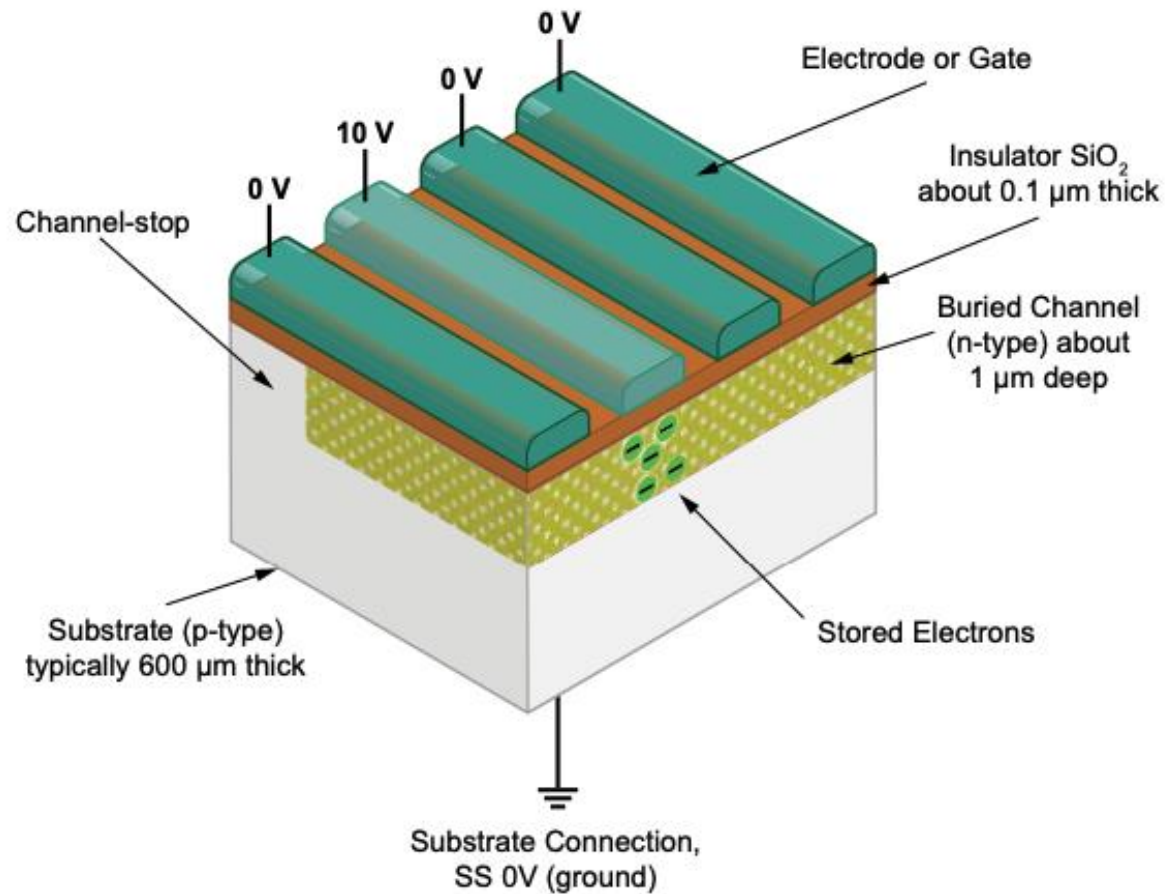
Single CCD Pixel



- The **band gap energy** is the gap between the valence and conduction band.

Detecting and Recording Light: CCD pixels

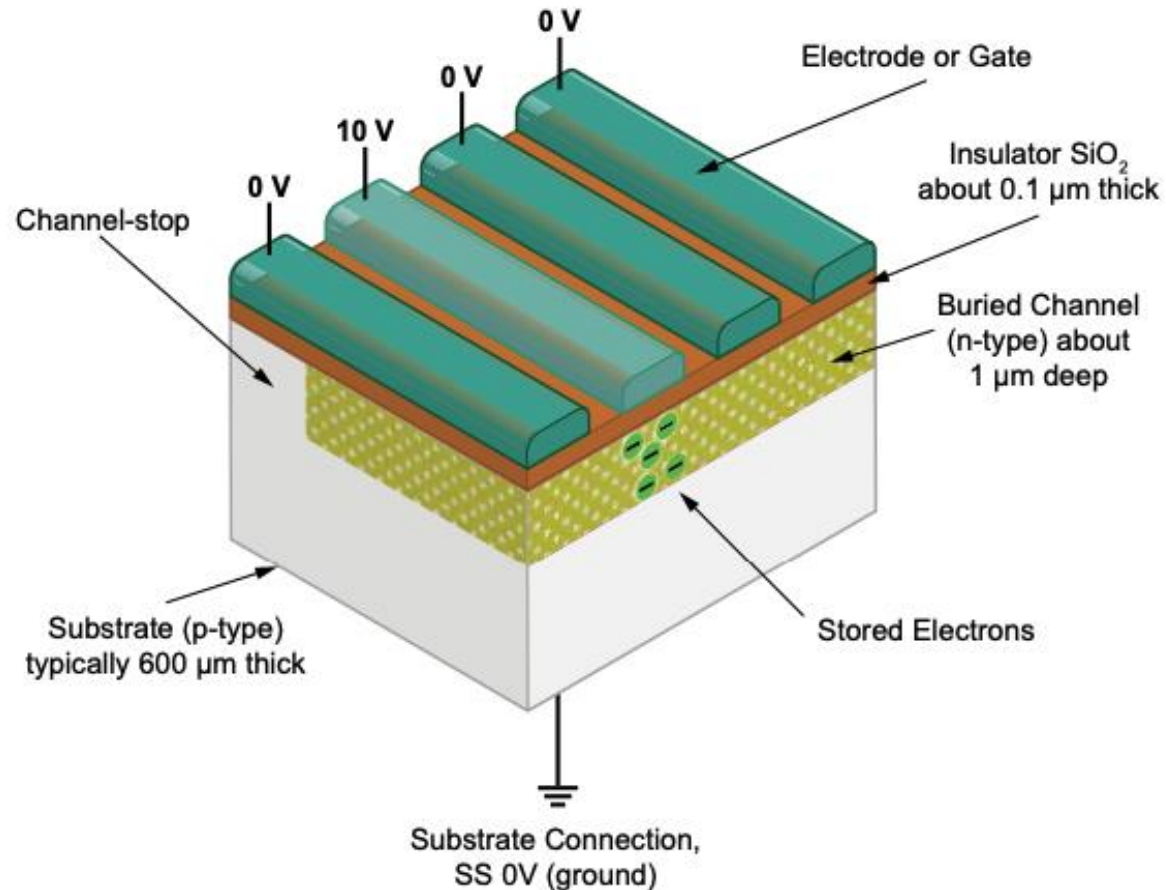
Single CCD Pixel



- The band gap energy (1.11 eV) for silicon is high enough where it is primarily sensitive to visible wavelengths.

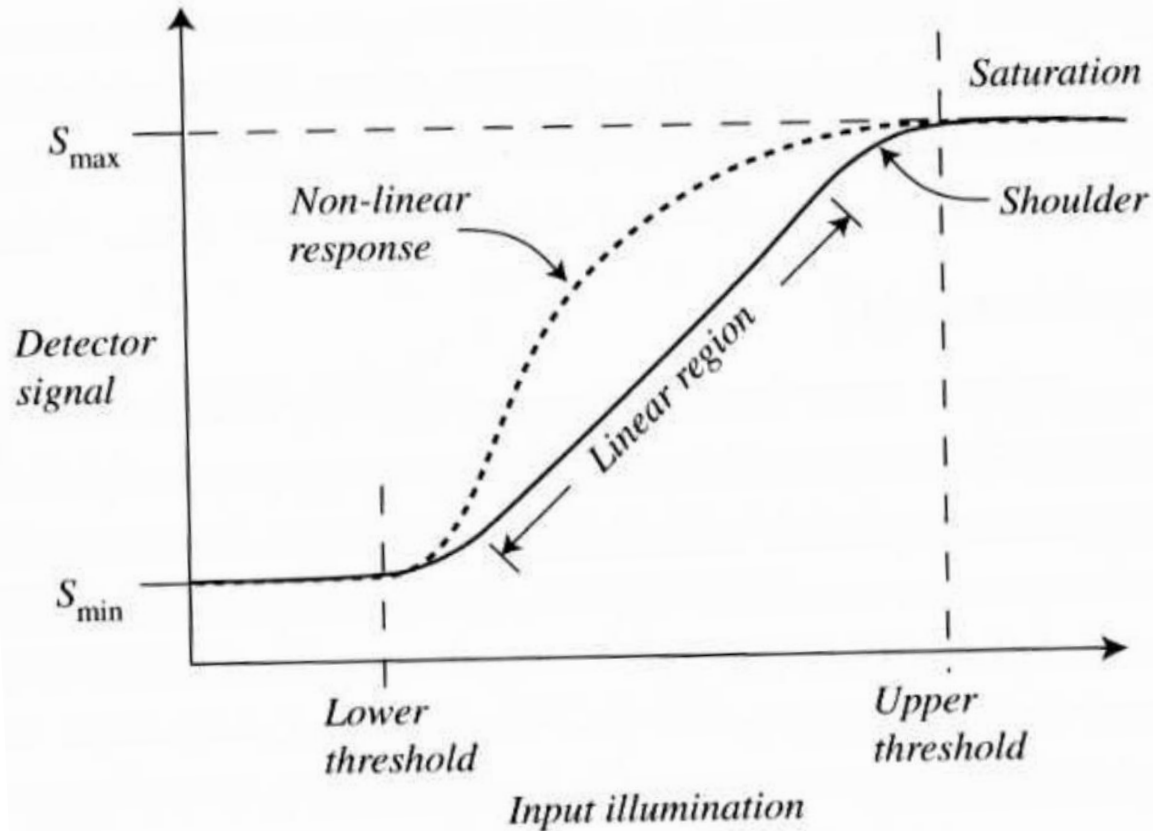
Detecting and Recording Light: CCD pixels

Single CCD Pixel



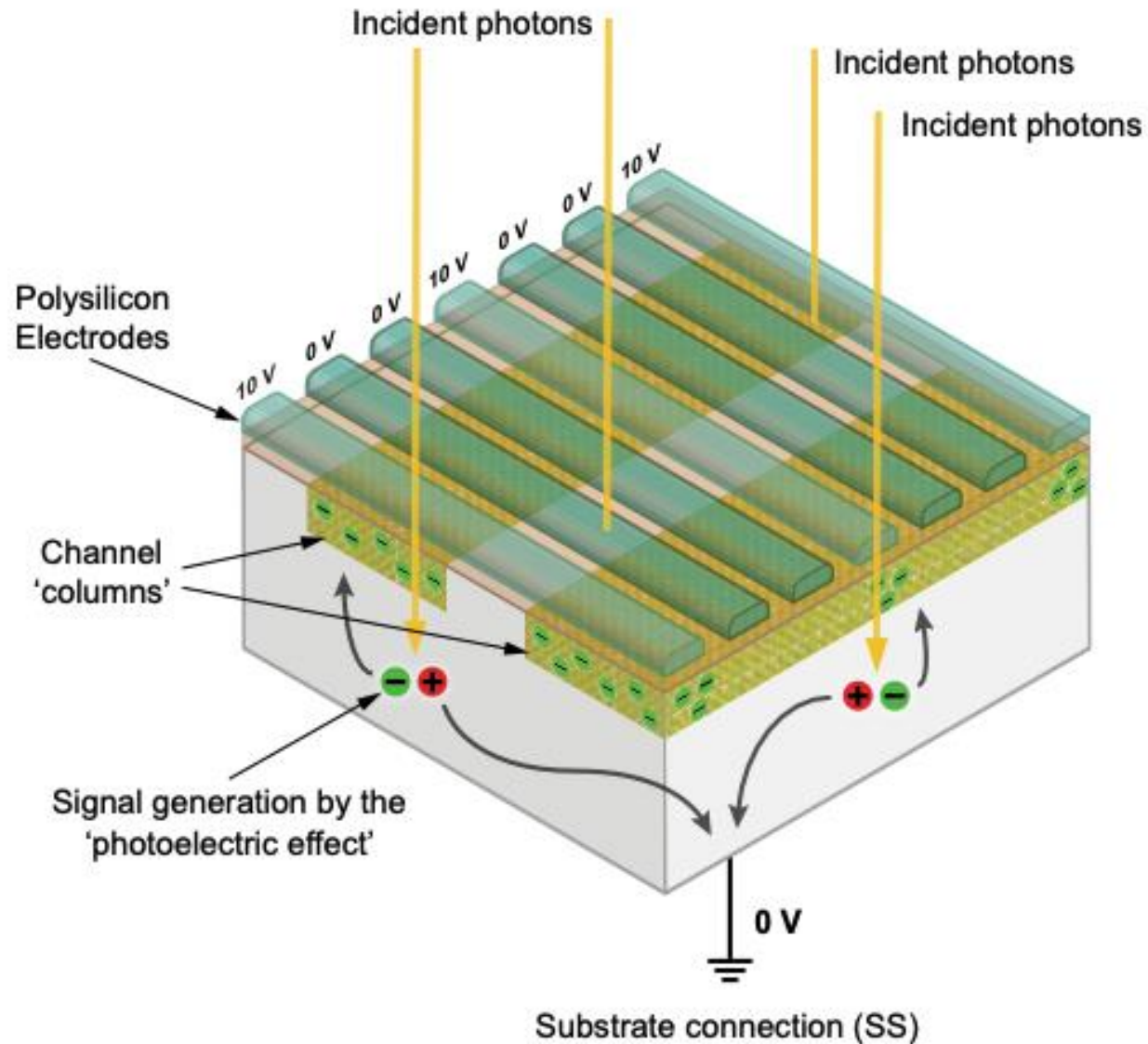
- If too many electrons accumulate within a pixel, it becomes **saturated**.
- At saturation, the electric field of the stored electrons starts to impact the pixel's response. The impact is often non-linear.
- There are ways to mitigate saturation when reducing data, but this is challenging to do unless your detector was used to study several sources of varying brightness.

Detecting and Recording Light: CCD pixels

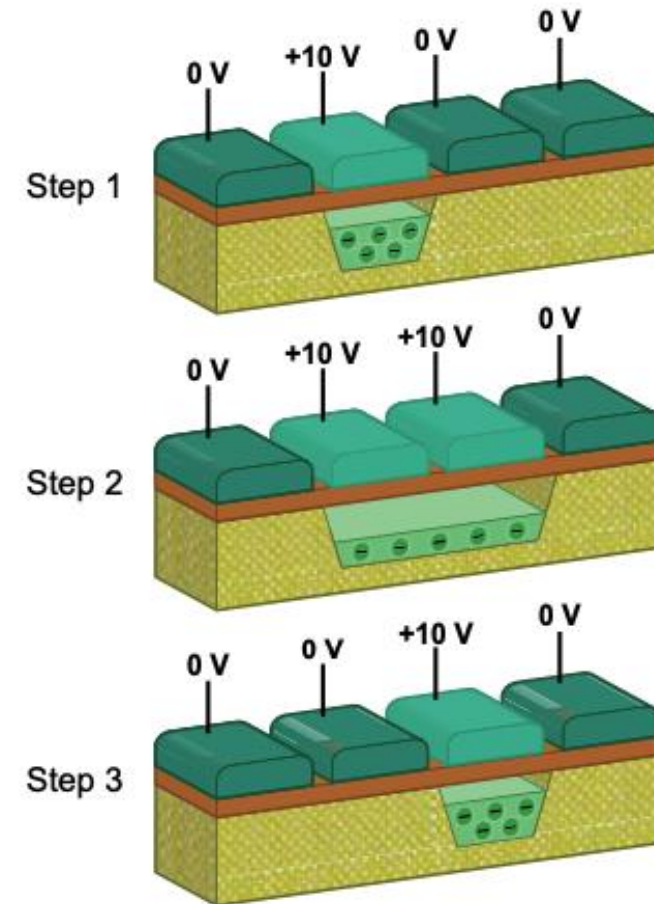


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Section of a CCD Array with Electrodes and Channel 'Columns'



Principle of Charge Transfer



- Voltages can be cycled to move charge along a column within potential wells.

Reading Out CCDs

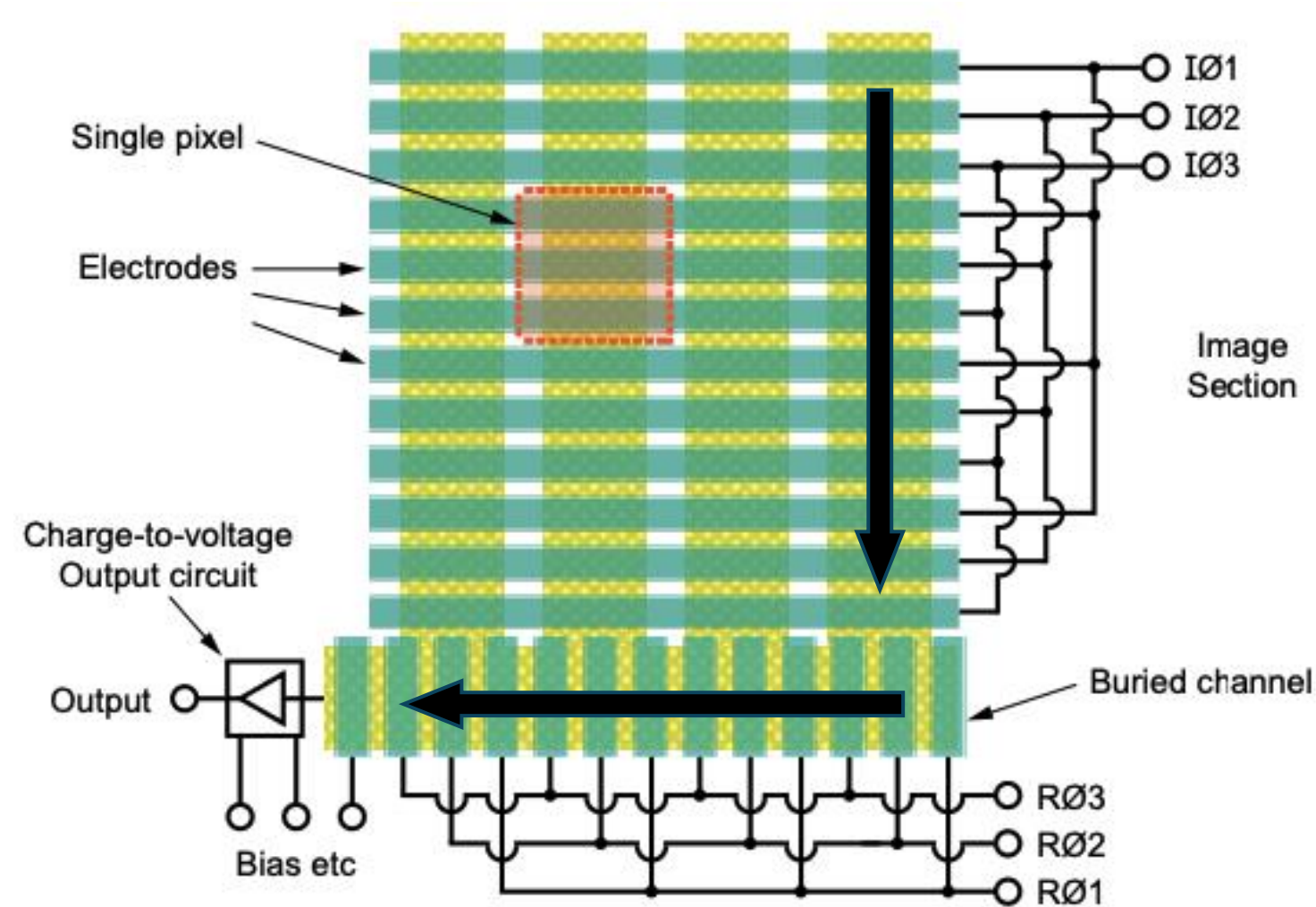
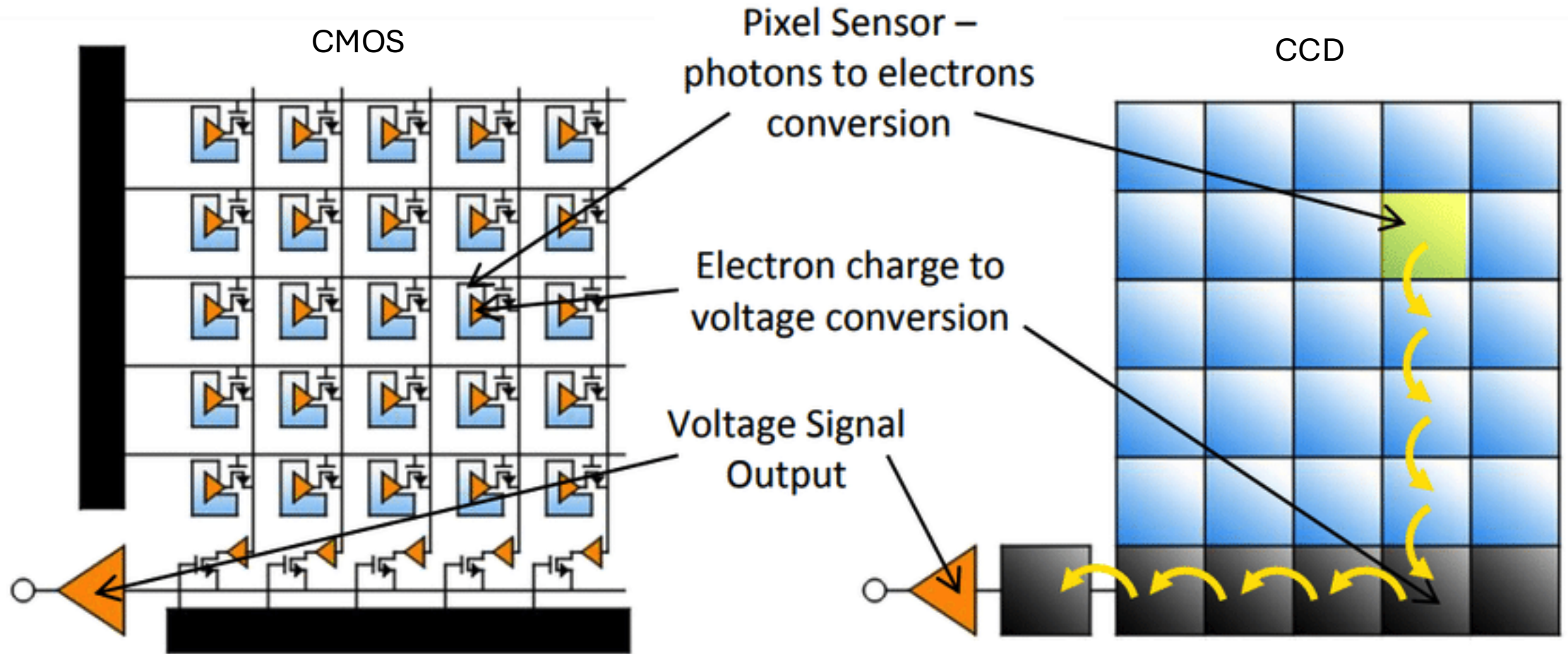


Image from Teledyne Imaging

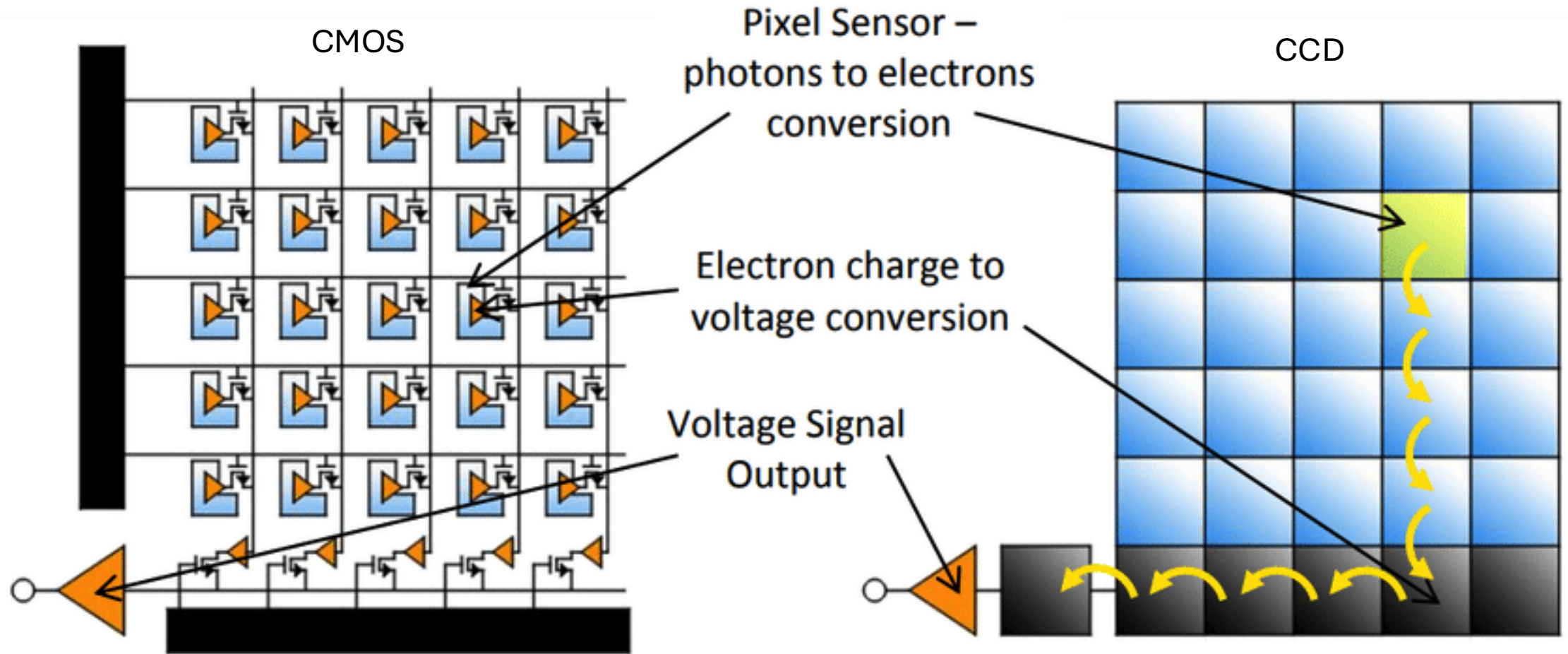
- Electrons are transferred down columns then each row of the columns is read from the readout register.
- The charge transfer process is very efficient but not perfect. This may impact very large format CCDs.
 - Example: 99.9996 % CTE
 - Detector Width: 4000
 - $0.999996^{(4000)} = 0.98$
- Electrons are converted to voltage at the output.

CMOS Read Out vs CCD Read Out



CMOS (Complementary Metal-Oxide-Semiconductor) detectors are used in your phones and other industrial applications. Some astronomical detectors use CMOS or CMOS-like readouts. Supposedly ~1000 times faster (Teledyne Vision Systems)

Detector Gain: Converting Voltages to Counts



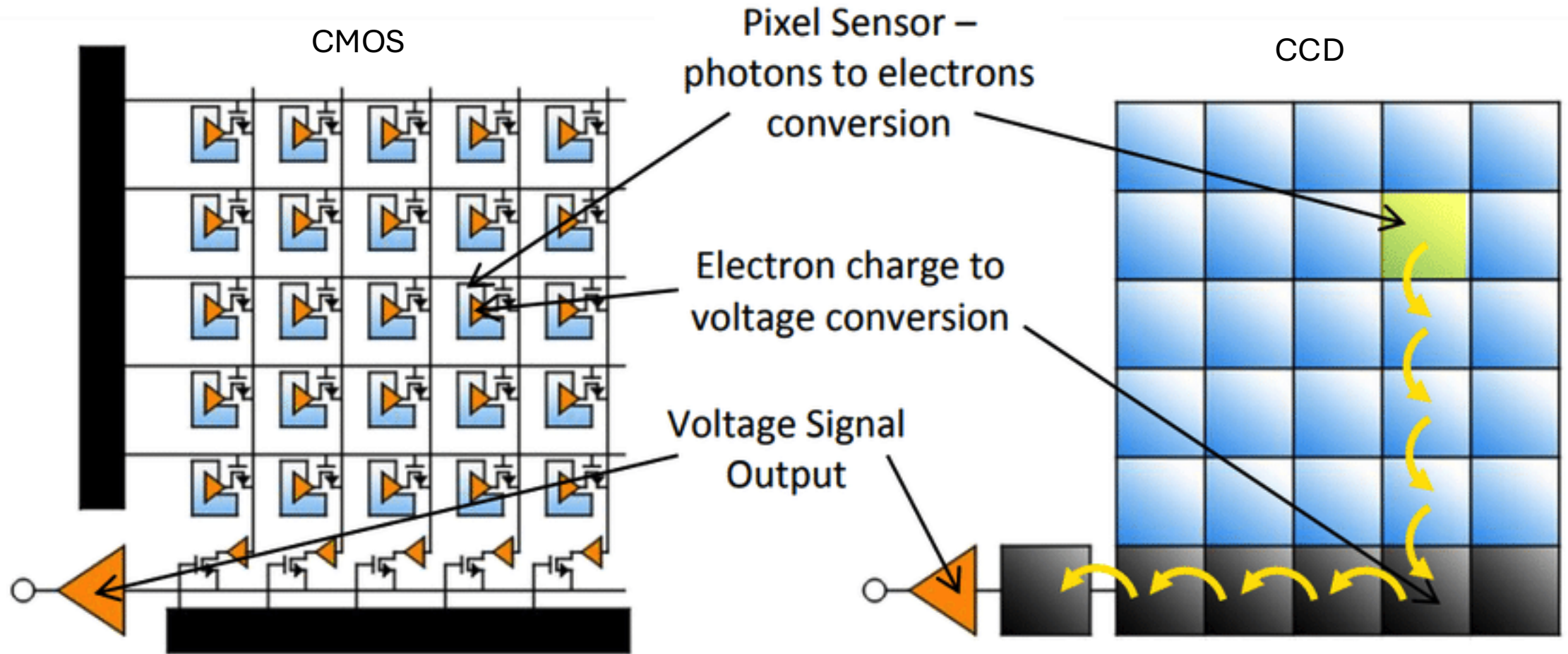
Analog to digital converters (ADC) converts analog voltage signals into discrete counts or Analog-to-Digital Units (ADU). This conversion is set by the gain (e^-/ADU).

ASTR 302 Fall 2026

Detectors Continued



Detector Gain: Converting Voltages to Counts



Analog to digital converters (ADC) converts analog voltage signals into discrete counts or Analog-to-Digital Units (ADU). This conversion is set by the gain (e^-/ADU).

Bits and Choosing ADC Gain



The amount of discrete steps your ADC has is determined by how many bits it has/reads.

A bit is a binary digit that communicates information as 0 or 1.

$$\text{Level \#} = 2^{\text{bits}}$$

Bits and Choosing ADC Gain



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A bit is a binary digit that communicates information as 0 or 1.

Level # = 2^{bits}

Bits and Choosing ADC Gain

1-bit	0		1	
2-bits	00 -> 0	01 -> 1	10 -> 2	11 -> 3
3-bits				
4-bits				
5-bits				
6-bits				
7-bits				
8-bits				
9-bits				
10-bits				
11-bits				
12-bits				
13-bits				
14-bits				

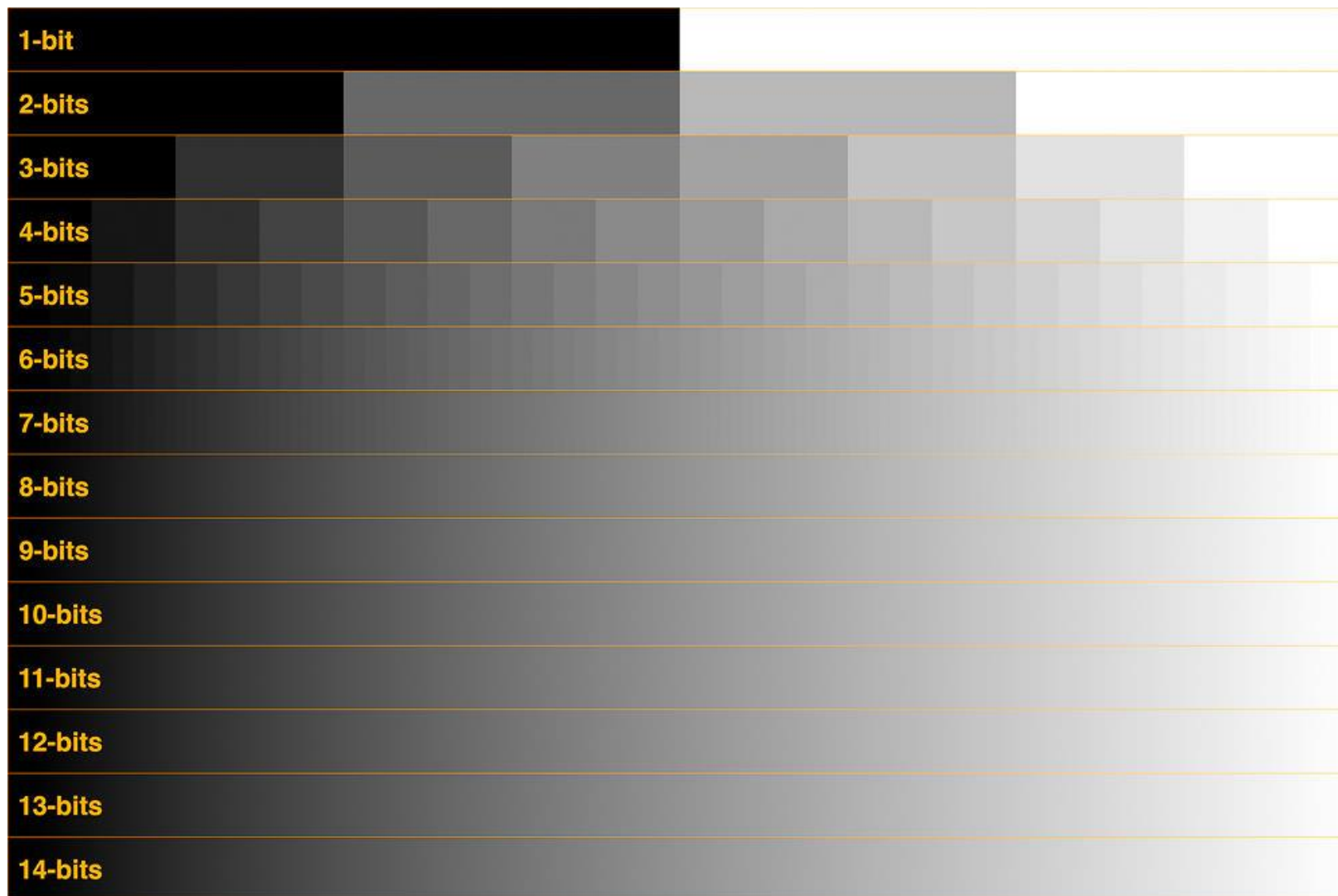
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Bits and Choosing ADC Gain

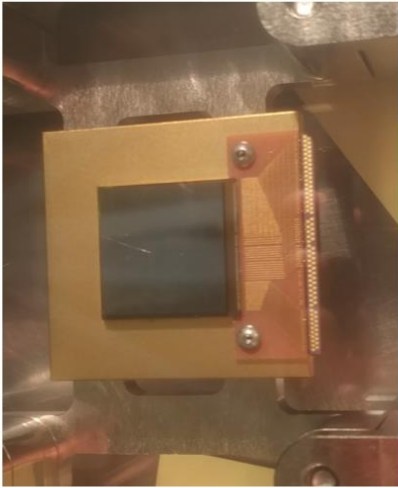
0 (e-) ← Well Depth (e-)



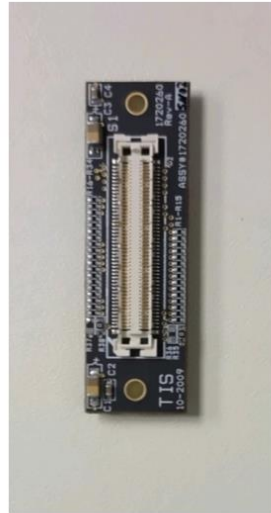
You usually set the gain to the ratio of the detector's well depth and the discrete voltage levels your ADC can measure.

$$g = \frac{\text{well depth (e-)}}{2^{\text{bits}}(\text{ADU})}$$

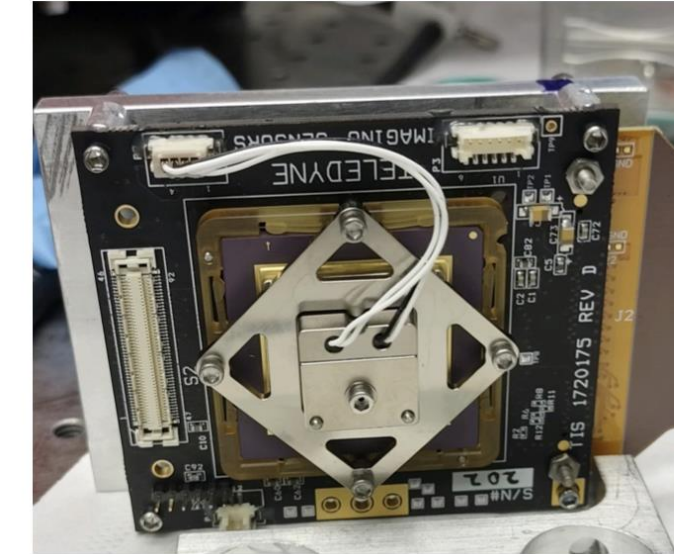
Read Out Electronics Example



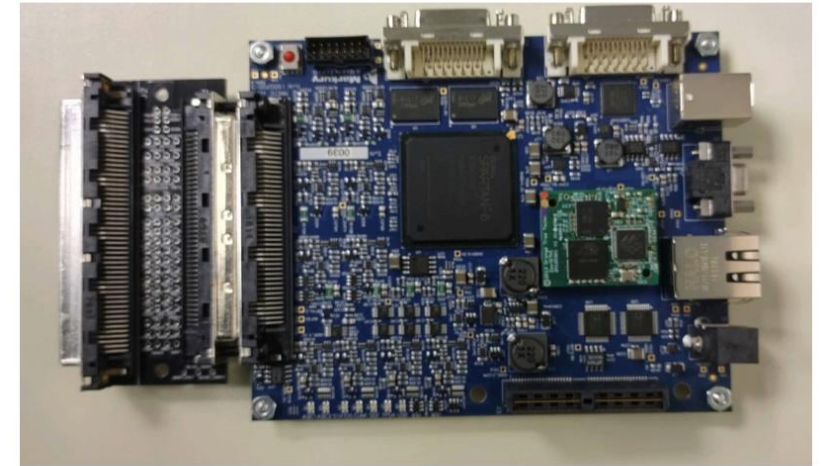
Detector



**Buffer
Board**



SIDECAR ASIC

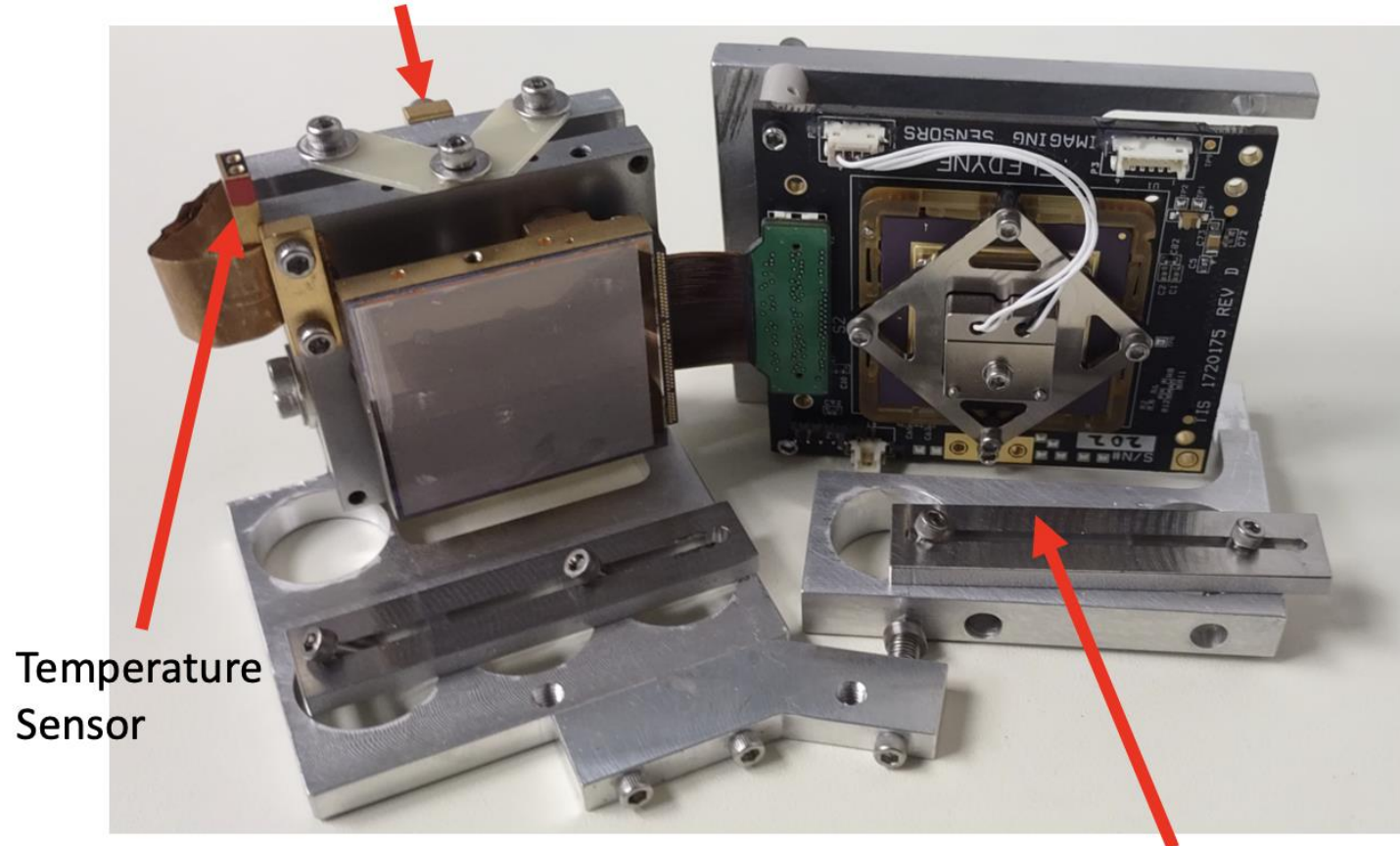


MACIE Card

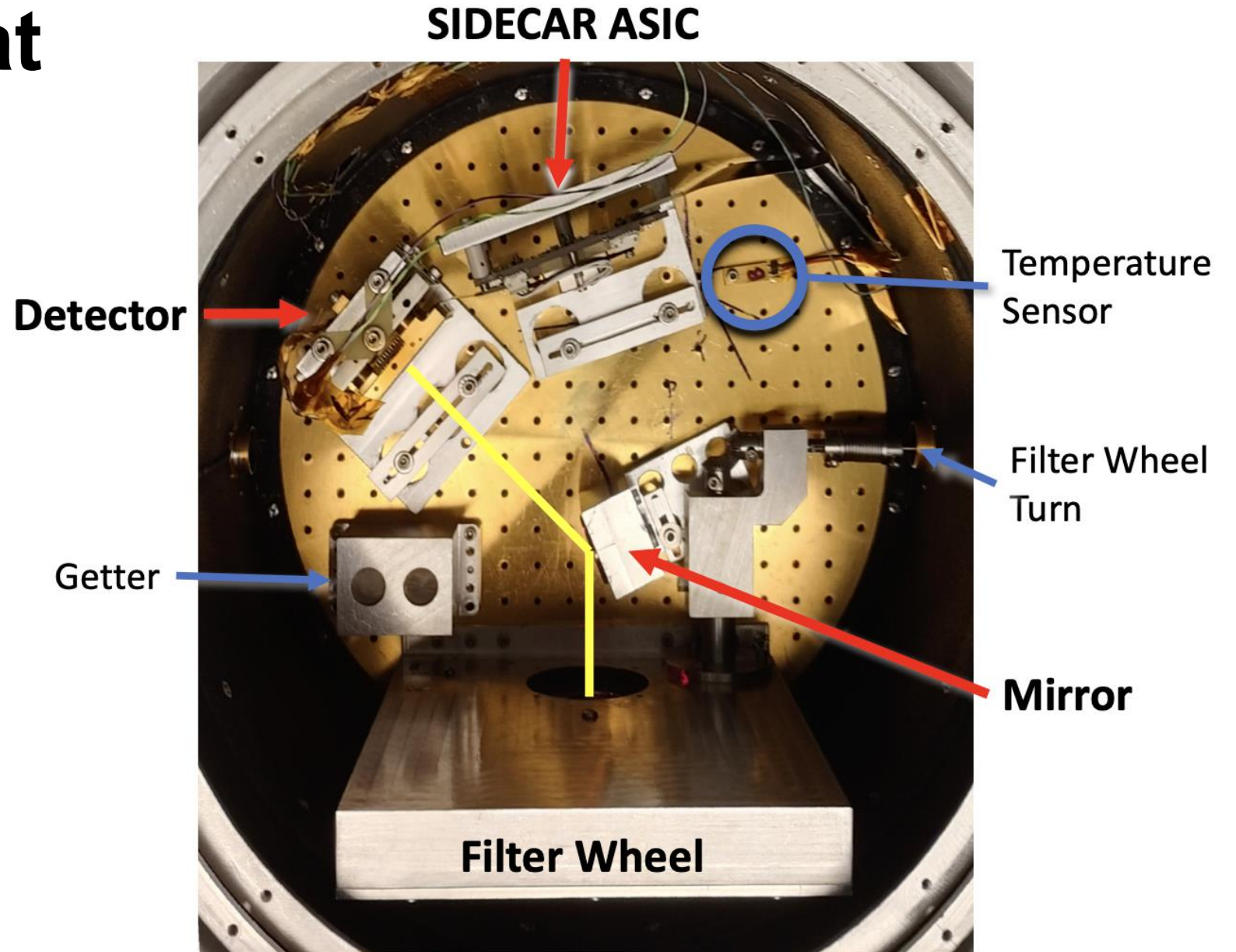
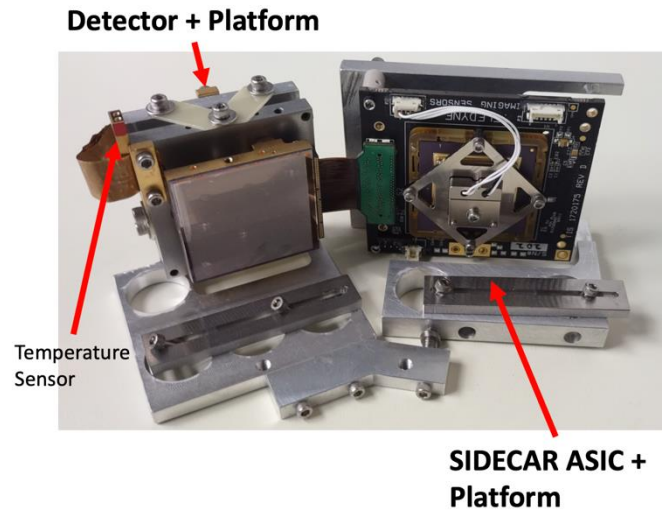
NOT A CCD!!!!!!!!!!!!!!
Hybrid CMOS

Read Out Electronics Example: Cryogenic Side

Detector + Platform



Inside a Cryostat

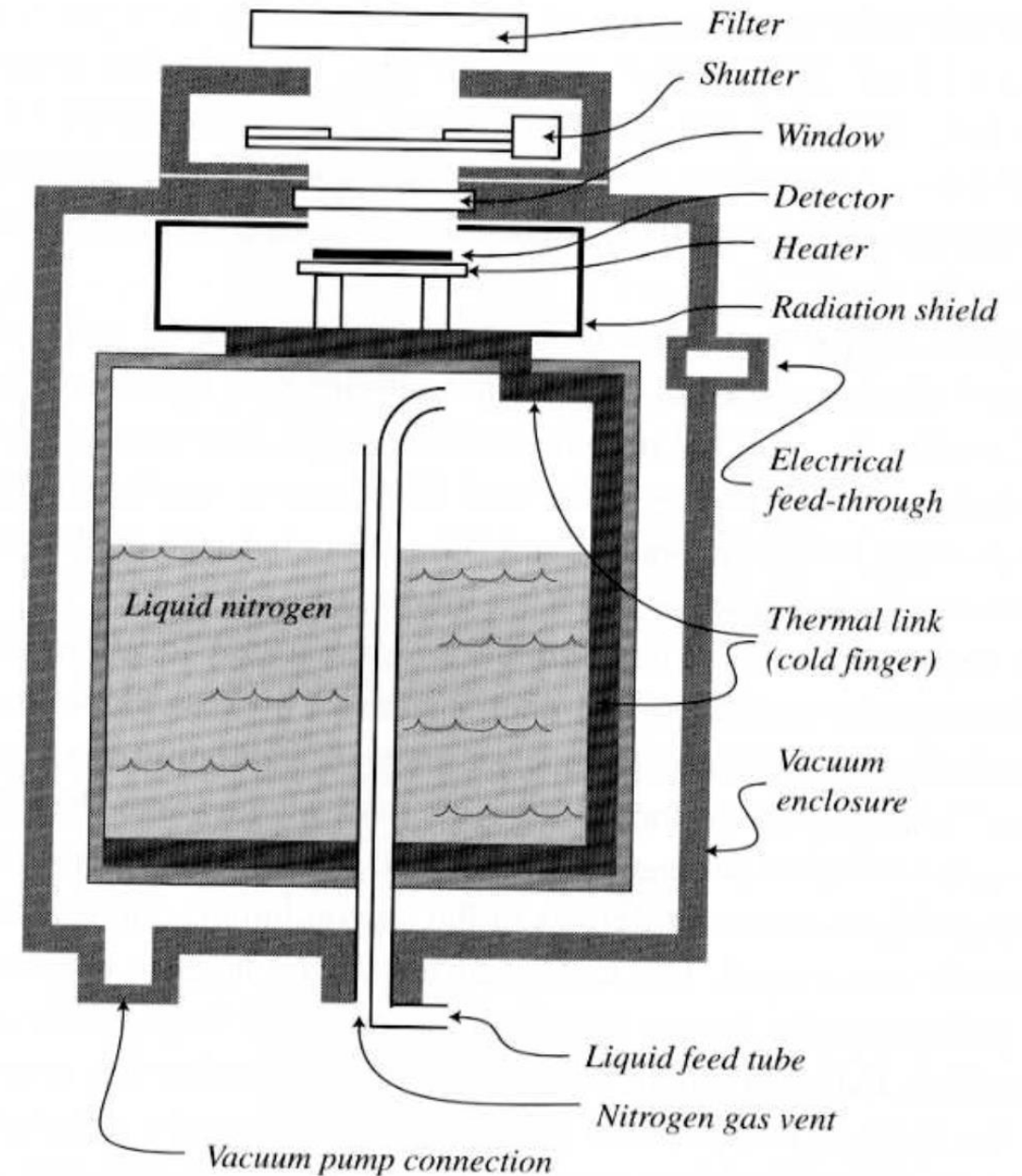


CCDs Need to Operate Cryogenically

The bandgap energy of Silicon is not hard to breach. You need to cool the CCDs itself to prevent the silicon from self emitting.

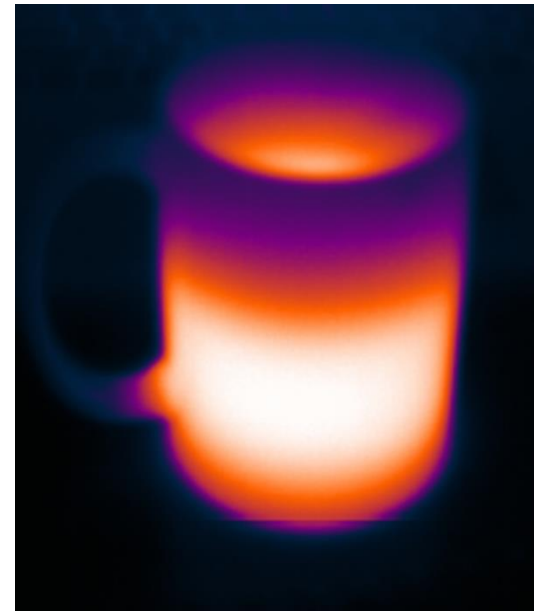
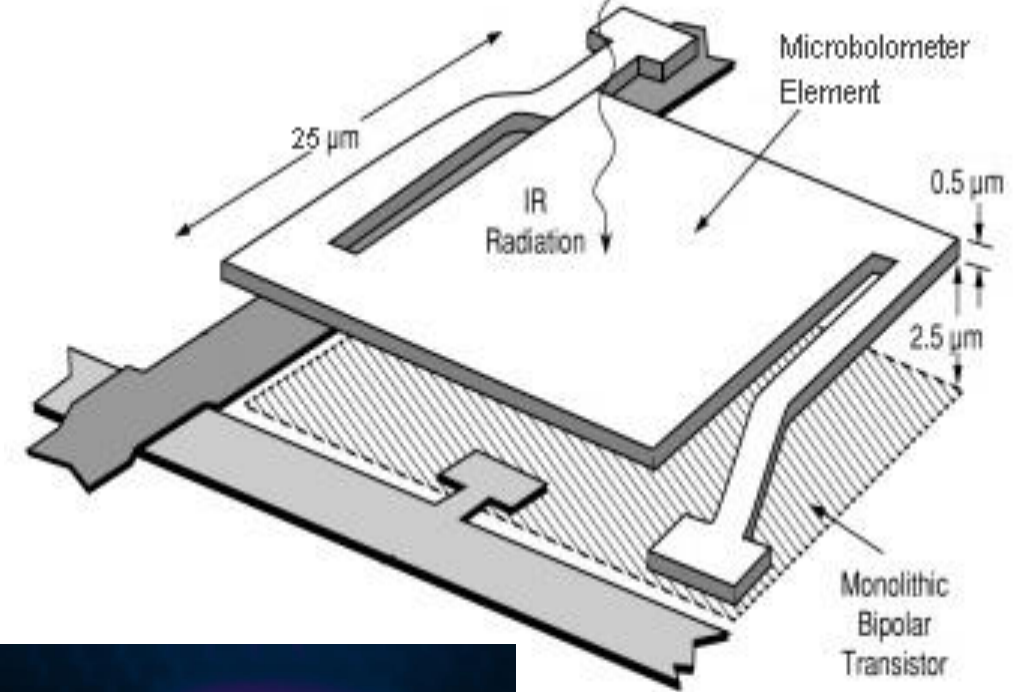
We cool CCDs in cryostats/dewars to reduce **dark current**.

For detectors sensitive in the infrared you need to cool the detector to limit dark current and emission from the instrument.



Thermal Detectors: Example Bolometers

- Thermal detectors measure changes in resistance to estimate the radiant heat of a source.
- A common IR sensitive material used for microbolometers is vanadium oxide (VOx) (8 – 14 micron sensitivity).
- They typically do not require cooling like CCDs or CMOS detectors, so they lack precision.
- Common uses: Firefighting, Night time studies of animals.

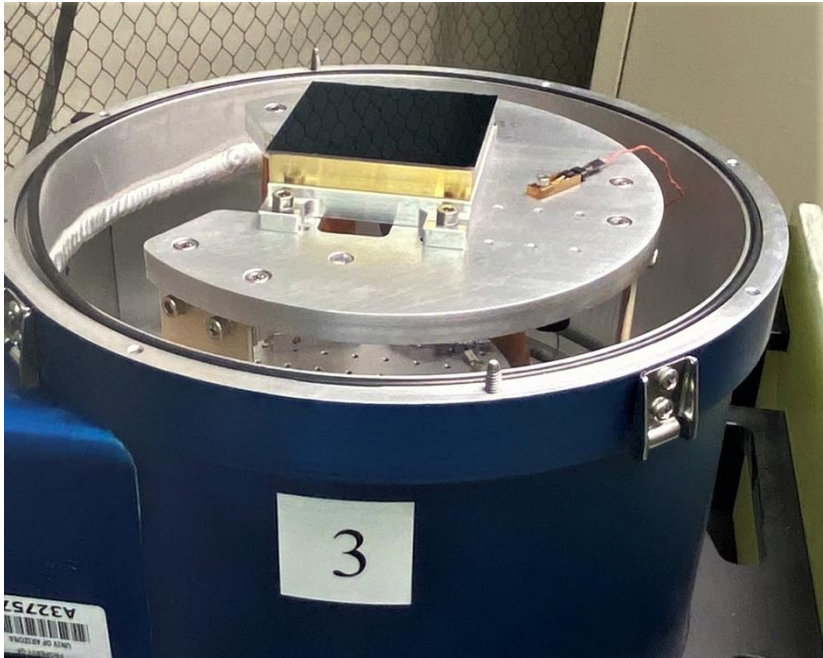


Noise Sources in CCDs

CCDs introduce both **random** and **systematic** noise into our observations. Therefore, CCDs impact both the precision and accuracy of our measurements.

To mitigate this we must characterize and understand several important properties:

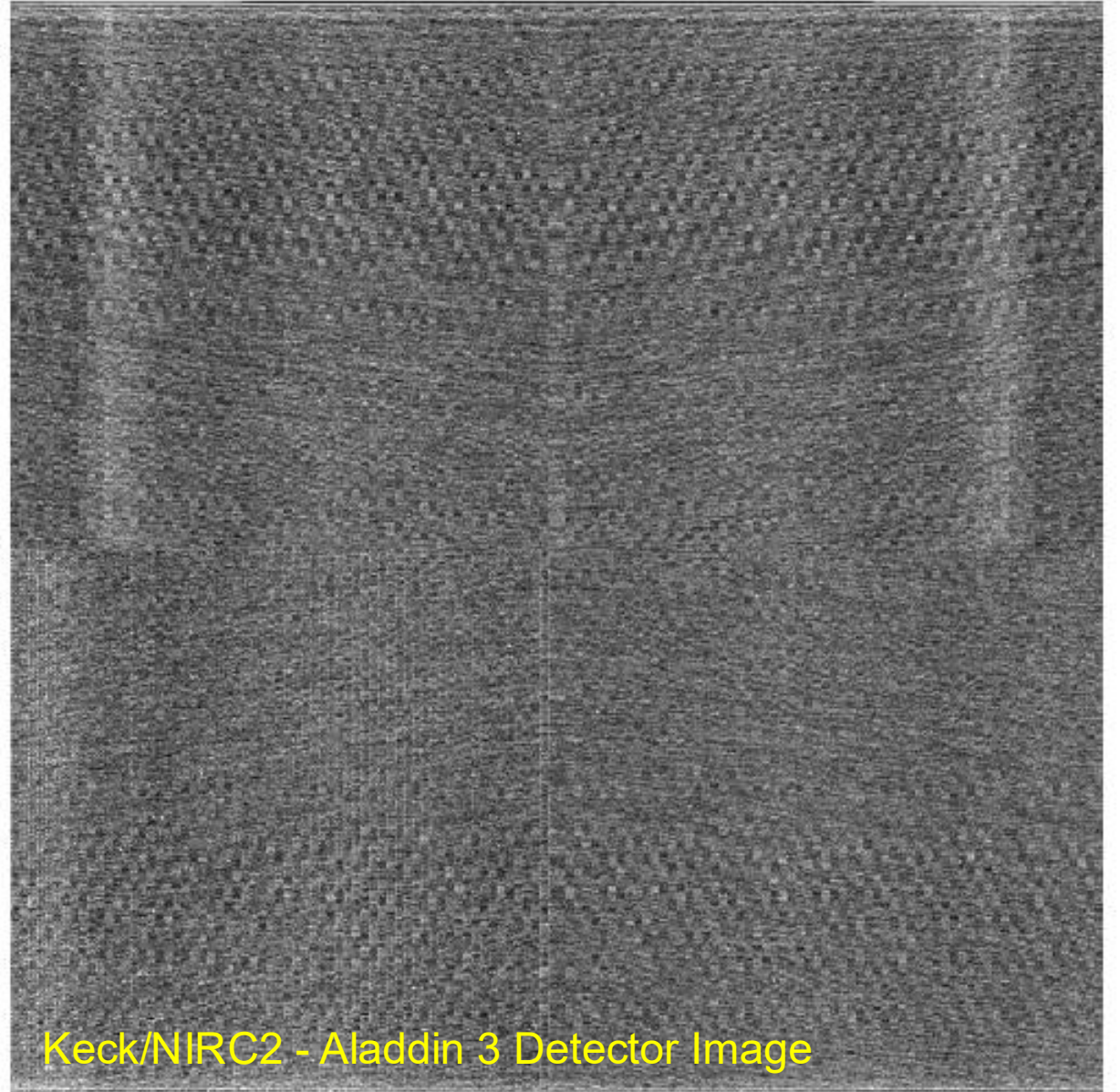
- **Bias** – Electronic Noise Offset
- **Dark Current** – Thermal Noise Offset
- **Flat Field** – Per pixel response uniformity
- **Read/Poisson Noise** – Counting precision



90 prime CCD exposed in dewar/cryostat.

Bias Images

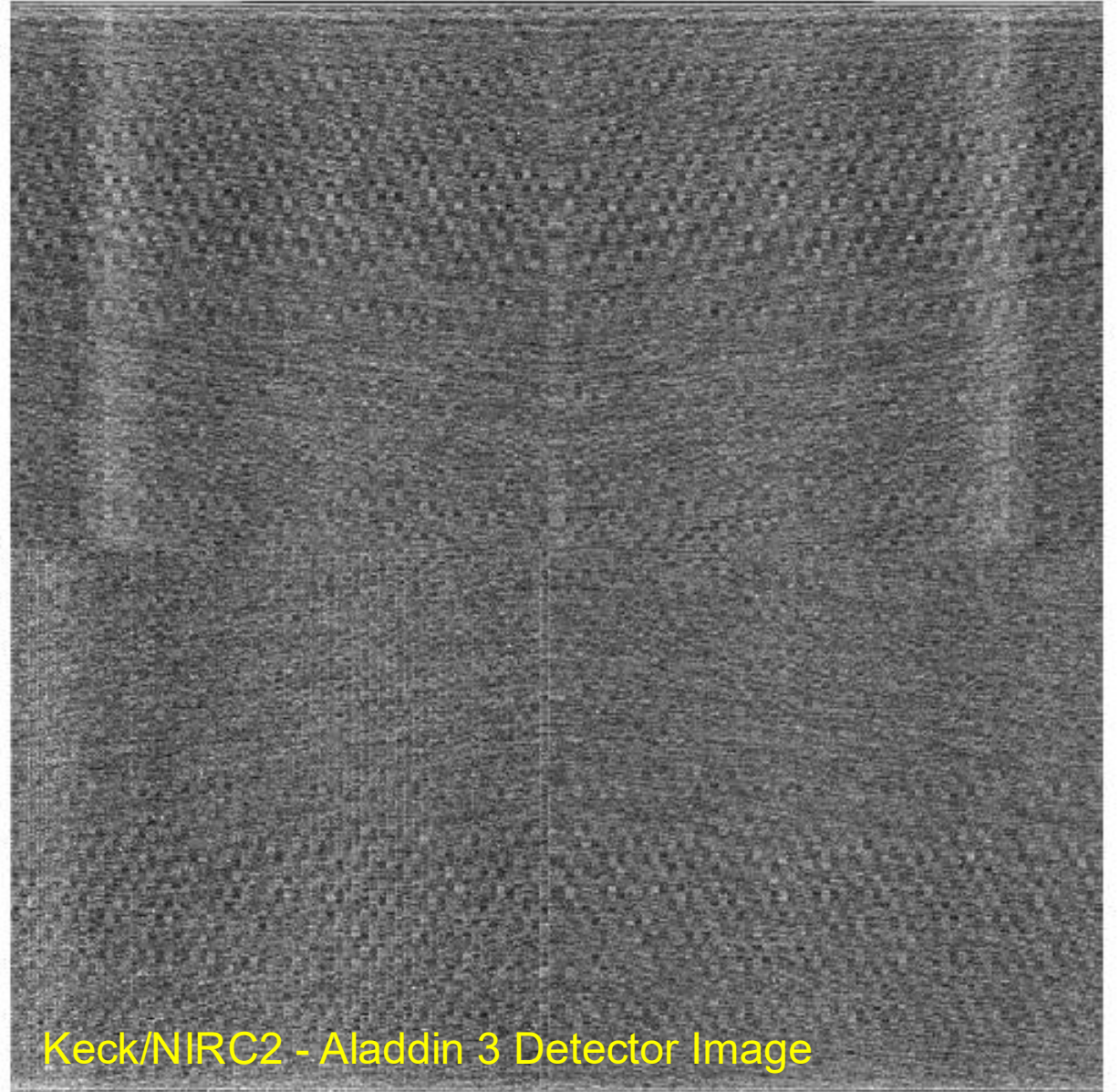
- A **bias image** is created by closing a shutter, resetting the charges on the CCD/detector and then reading it out with zero or very little exposure time.



Keck/NIRC2 - Aladdin 3 Detector Image

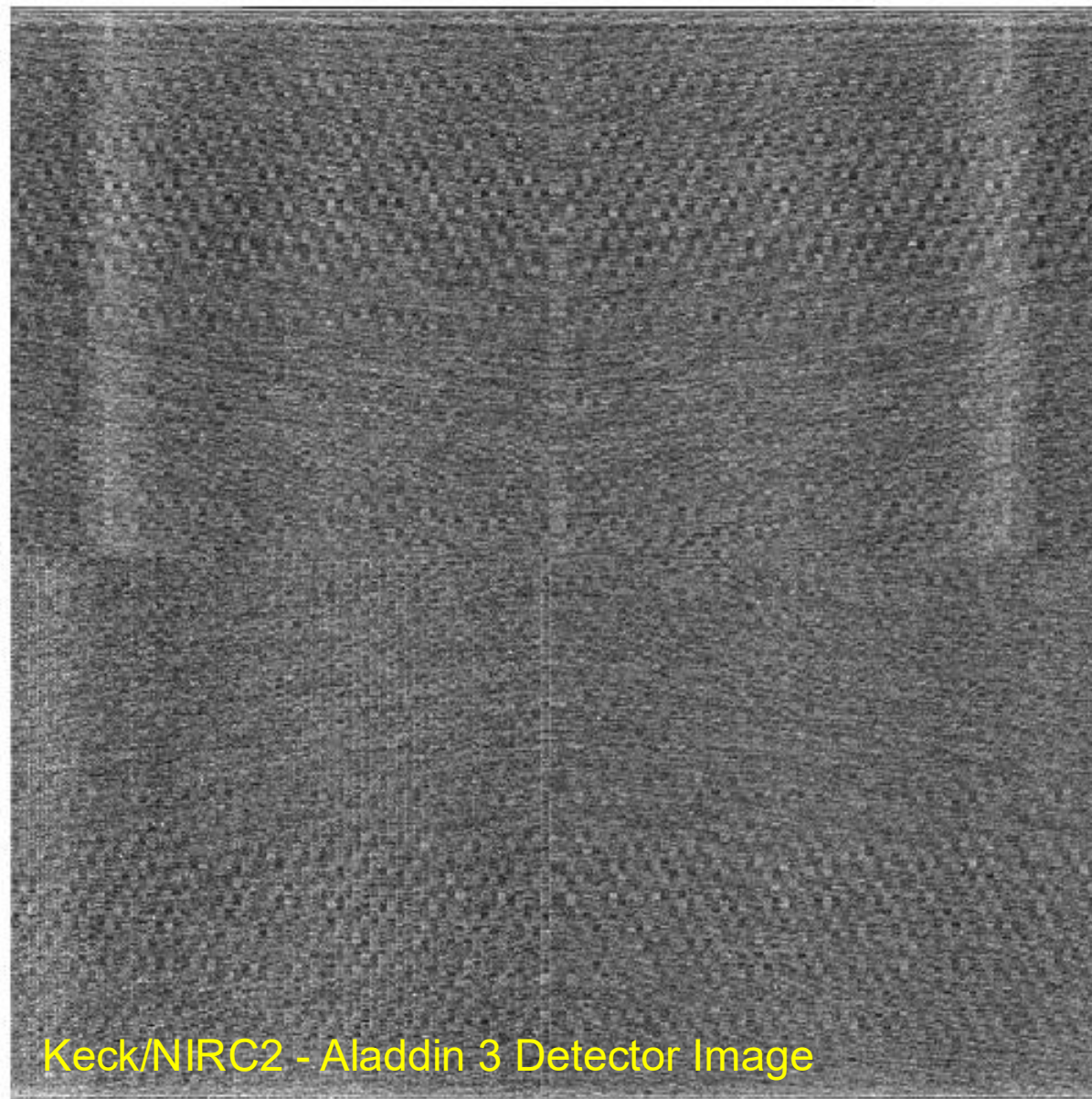
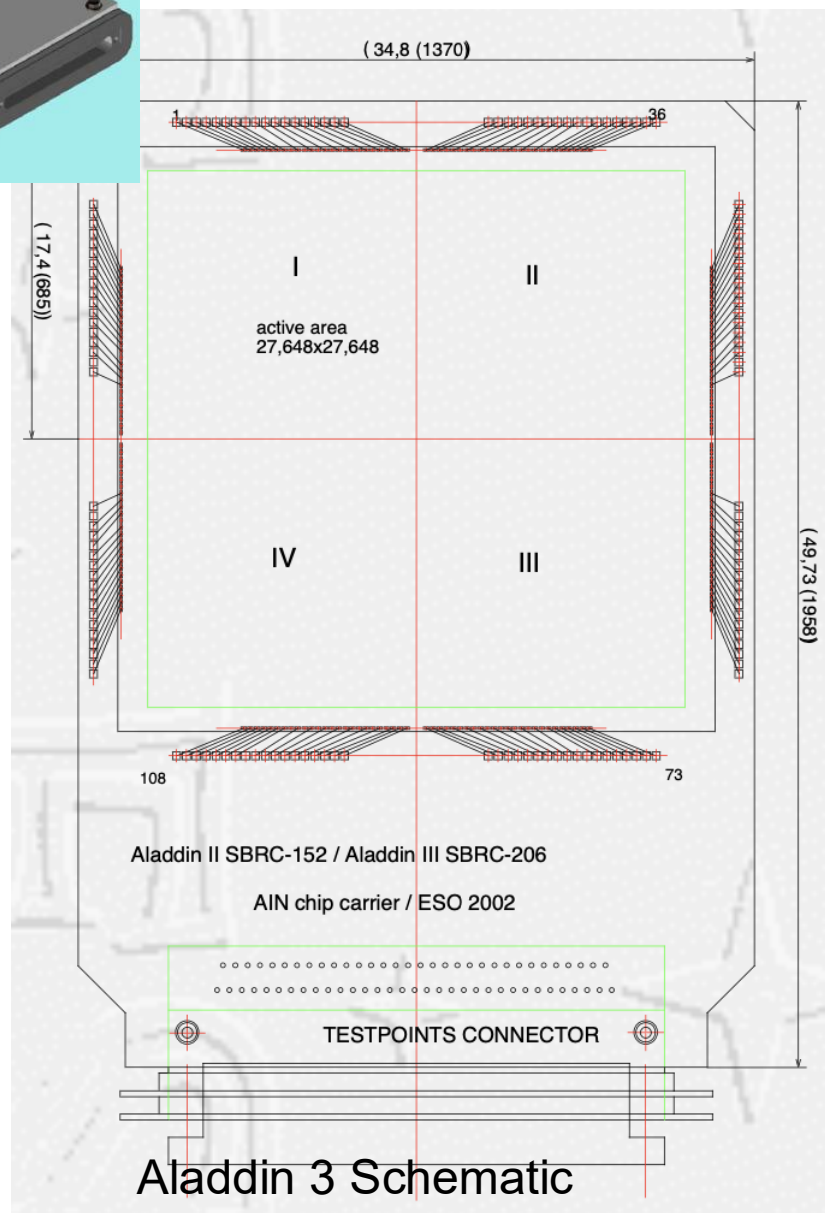
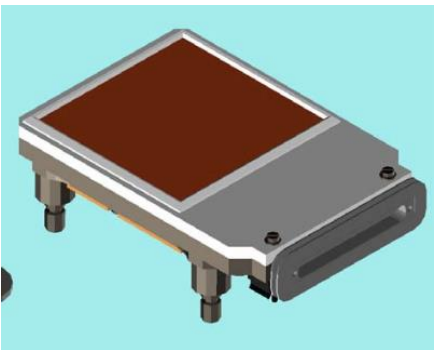
Bias Images

- A **bias image** is created by closing a shutter, resetting the charges on the CCD/detector and then reading it out with zero or very little exposure time.
- The bias image reflects the voltage offset applied by the read out electronics.

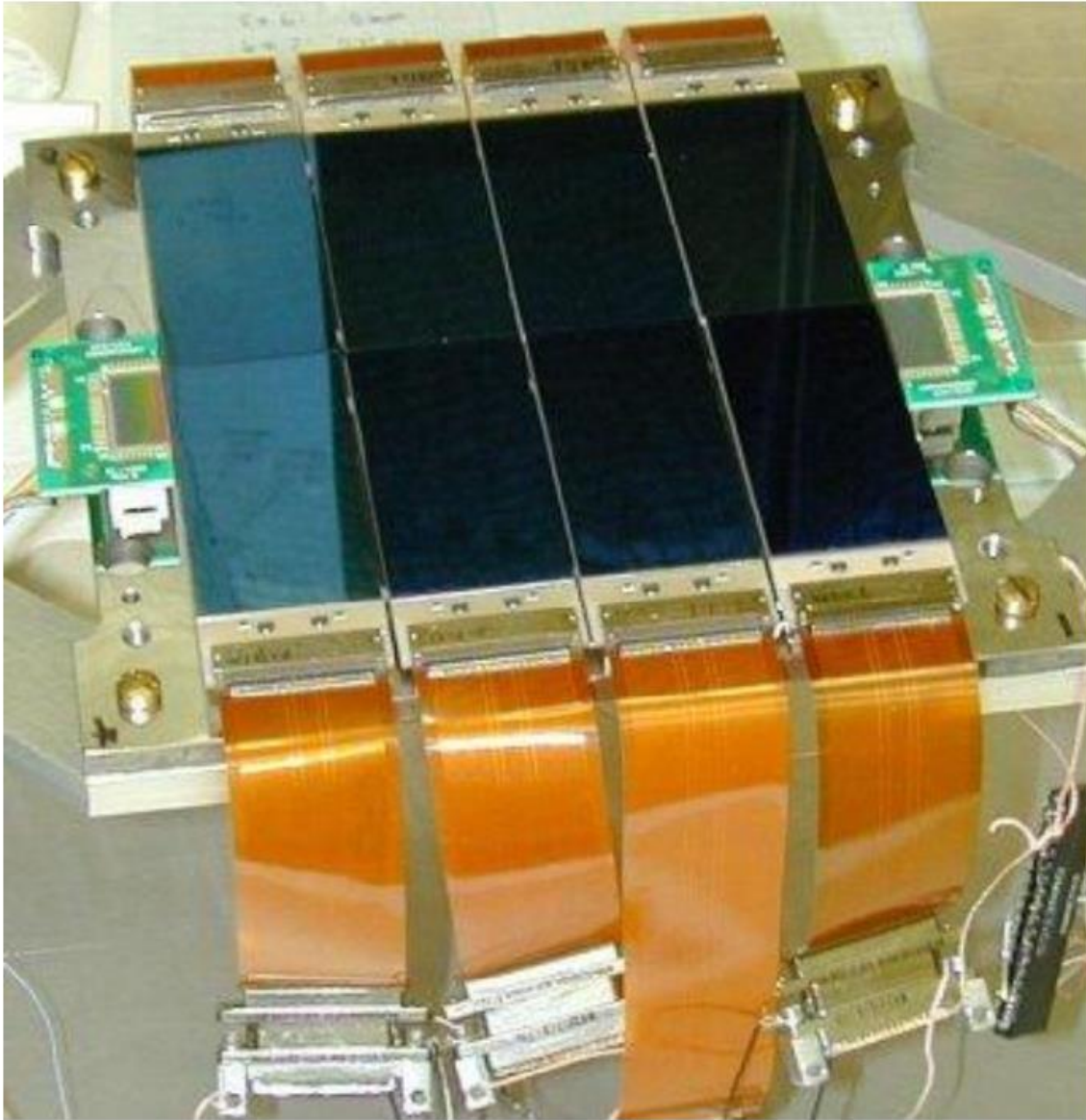


Keck/NIRC2 - Aladdin 3 Detector Image

Bias Images



Bias Images



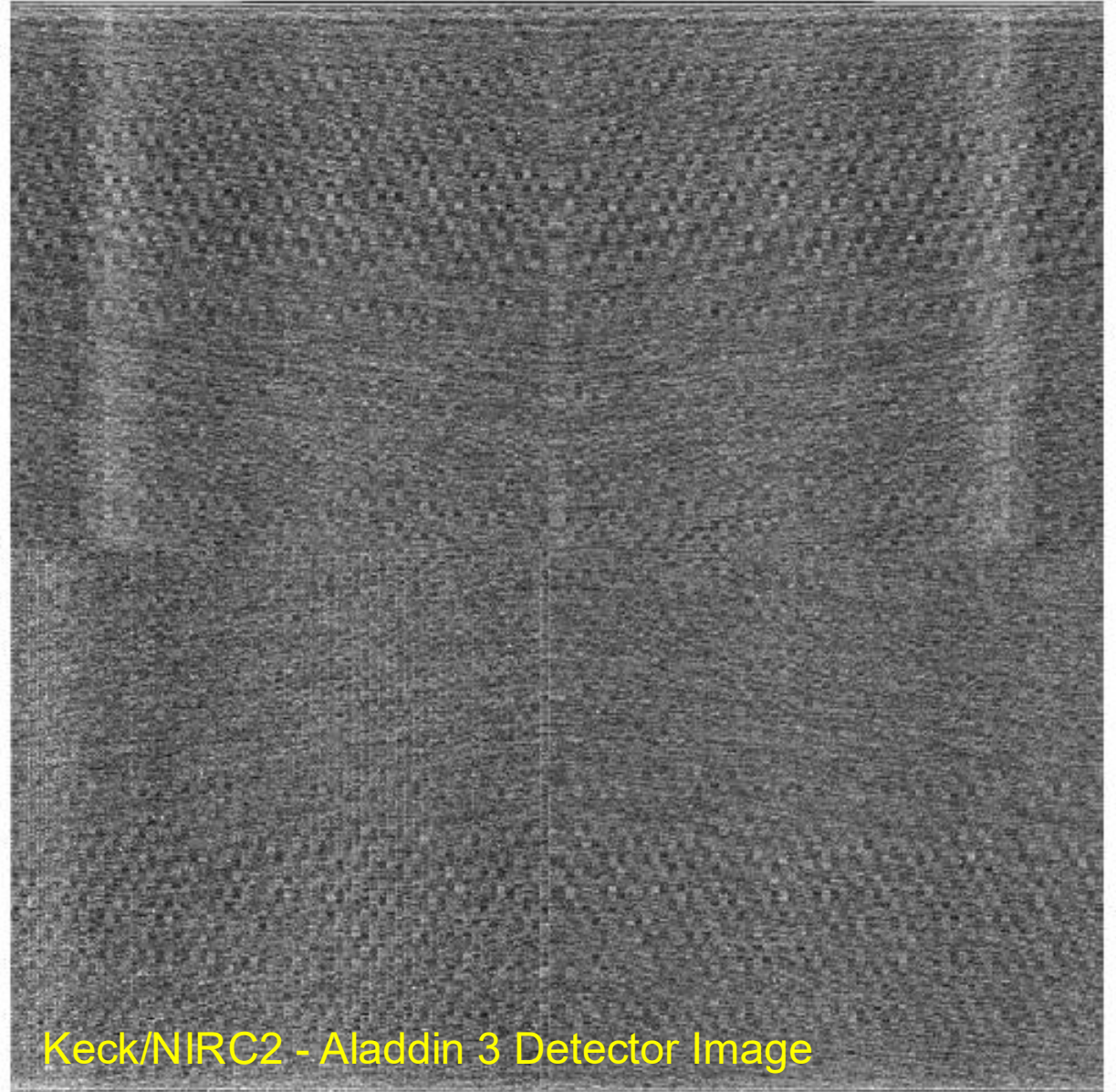
Keck/DEIMOS – CCD Array IRL



Keck/DEIMOS – CCD Array Image

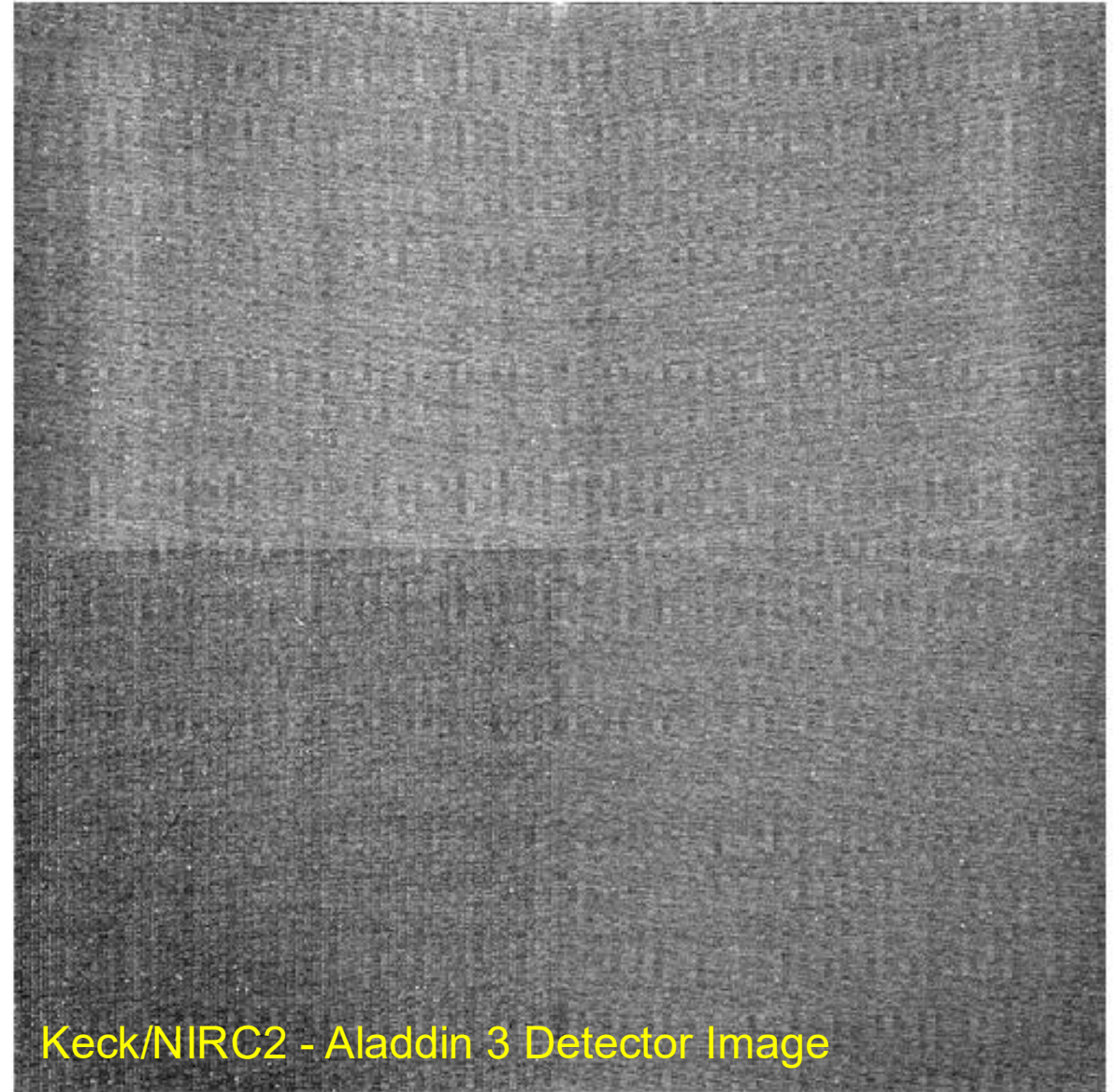
Bias Images

- A **bias image** is created by closing a shutter, resetting the charges on the CCD/detector and then reading it out with zero or very little exposure time.
- The bias image reflects the voltage offset applied by the read out electronics.
- Sometimes the bias can be estimated by **reference pixels** or an **overscan** region. Rows or columns of the detector that are not light sensitive.



Dark Images

- A **dark image** is created by closing a shutter, resetting the charges on the CCD/detector and then accumulating charge for some exposure time.
- Thermal heat of the detector/system causes electrons to collect on each pixel over time.
- Ideally you want your dark images to be the same exposure time as your science observations.



Flat Field Images

- A **flat field image** is created by exposing the CCD/detector to a uniformly illuminated source:
 - Dome flat (bright)
 - Sky (dim, has stars in it)
 - Source (bright)
- Each pixel on a detector has its own personality. If you uniformly illuminate a detector, the image or response would not be uniform.
- The normalized flat field image is used to adjust the response of your science images.



Reducing Your Science Observations

Step 1: Create normalized flat field image (F_n).

Step 2: (If necessary) scale dark image (D) to the same exposure time as your science image.

Step 3: Subtract the Bias image (B) and scaled dark image (sD) from your science exposure (S).

Step 4: Divide the adjusted science image by the normalized flat field to make a reduced scientific image.

Creating Normalized Flat Field (Fn)

$$Fn(x, y) = \frac{F(x,y) - B(x,y)}{\text{np.median}(F(x,y) - B(x,y))}$$

$$Fn(x, y) = \frac{\text{Median} \left(\begin{array}{c} \text{Image 1} - \text{Image 2} \\ \text{Image 1} - \text{Image 2} \end{array} \right)}{\text{Median} \left(\begin{array}{c} \text{Image 1} - \text{Image 2} \\ \text{Image 1} - \text{Image 2} \end{array} \right)}$$

The diagram illustrates the process of creating a normalized flat field (Fn) by subtracting a background image (B) from a flat field image (F) and then normalizing the result by the median of the difference. The visual representation shows two identical grayscale images, each with a dark spot in the bottom-left corner, being subtracted from each other to produce a flat field.

Creating Normalized Flat Field (Fn)

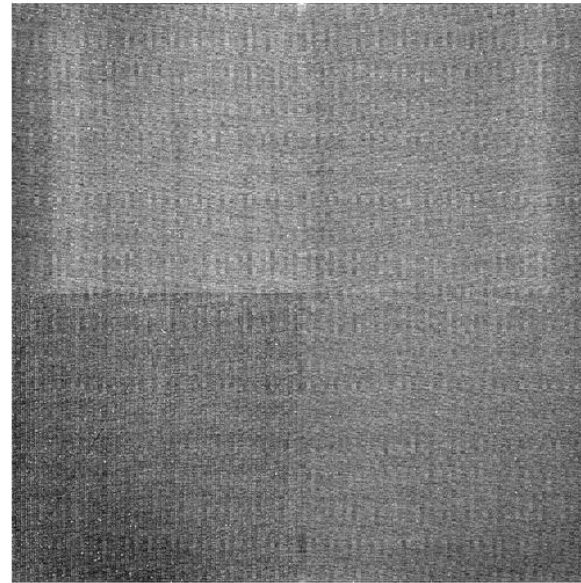
$$F_n(x, y) = \frac{F(x,y) - B(x,y)}{\text{np.median}(F(x,y) - B(x,y))}$$

Note: For CCDs, you absolutely need to remove the bias. For some detectors with non-destructive reads, you can estimate the pedestal or offset from “reading up the ramp”.

Creating Scaled Dark Image (sD)

$$\text{sD}(x, y) = \frac{t_s}{t_d} * D(x, y)$$

$$\text{sD}(x, y) = \frac{t_s}{t_d} *$$

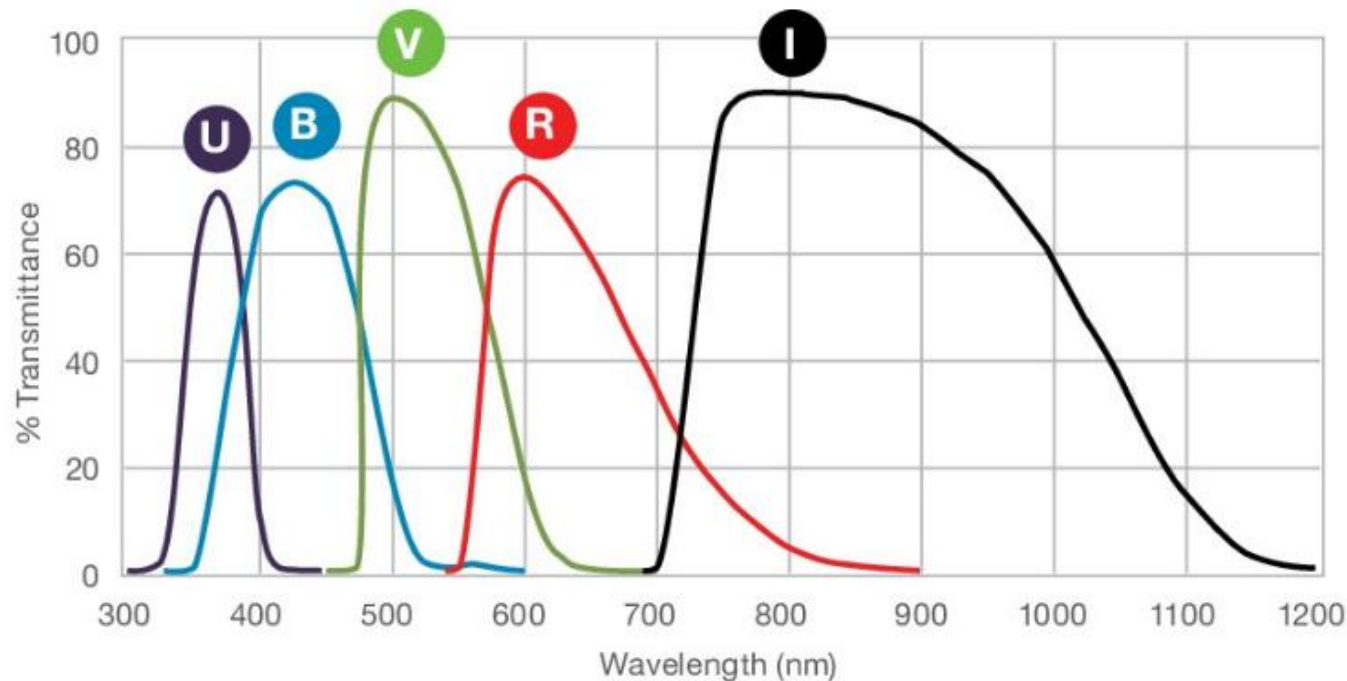


Creating a Reduced Scientific Image

$$\text{Reduced Image (x, y)} = \frac{S(x,y) - sD(x,y) - B(x,y)}{Fn(x,y)}$$

The reduced image has the systematic offsets of the detector removed. There are still steps to convert the data into physical units like flux.

Johnson/Bessel UBVRI Filters



- U filter (P/N JOHN-U-XX)
- B filter (P/N JOHN-B-XX)
- V filter (P/N JOHN-V-XX)
- R filter (P/N JOHN-R-XX)
- I filter (P/N JOHN-I-XX)

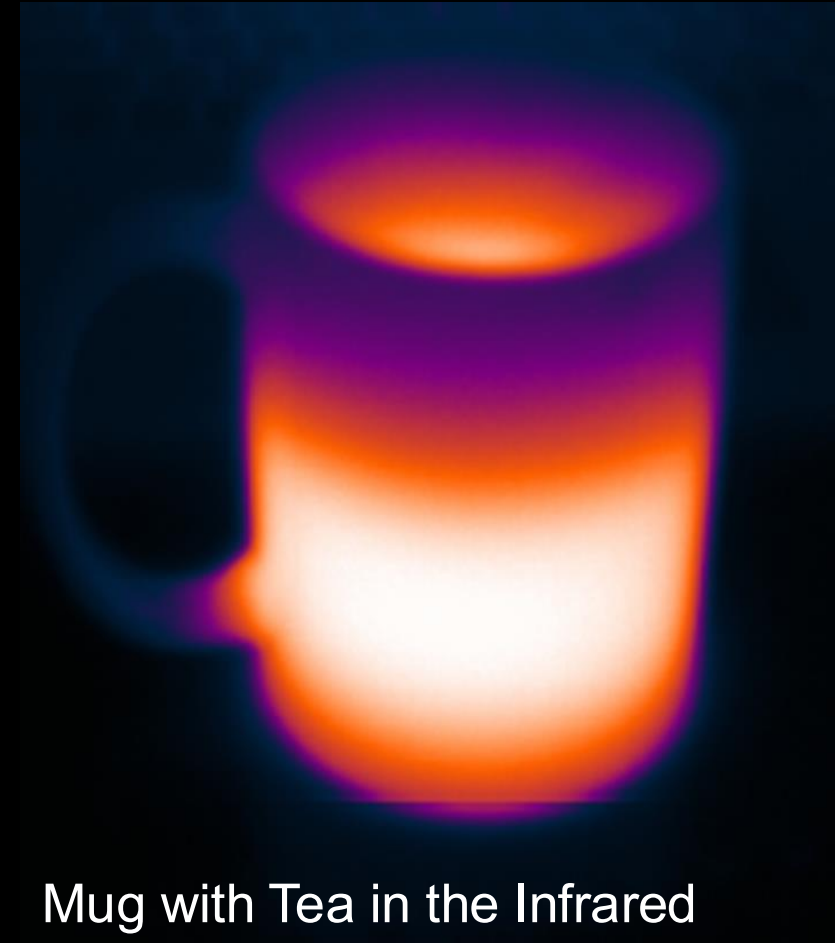
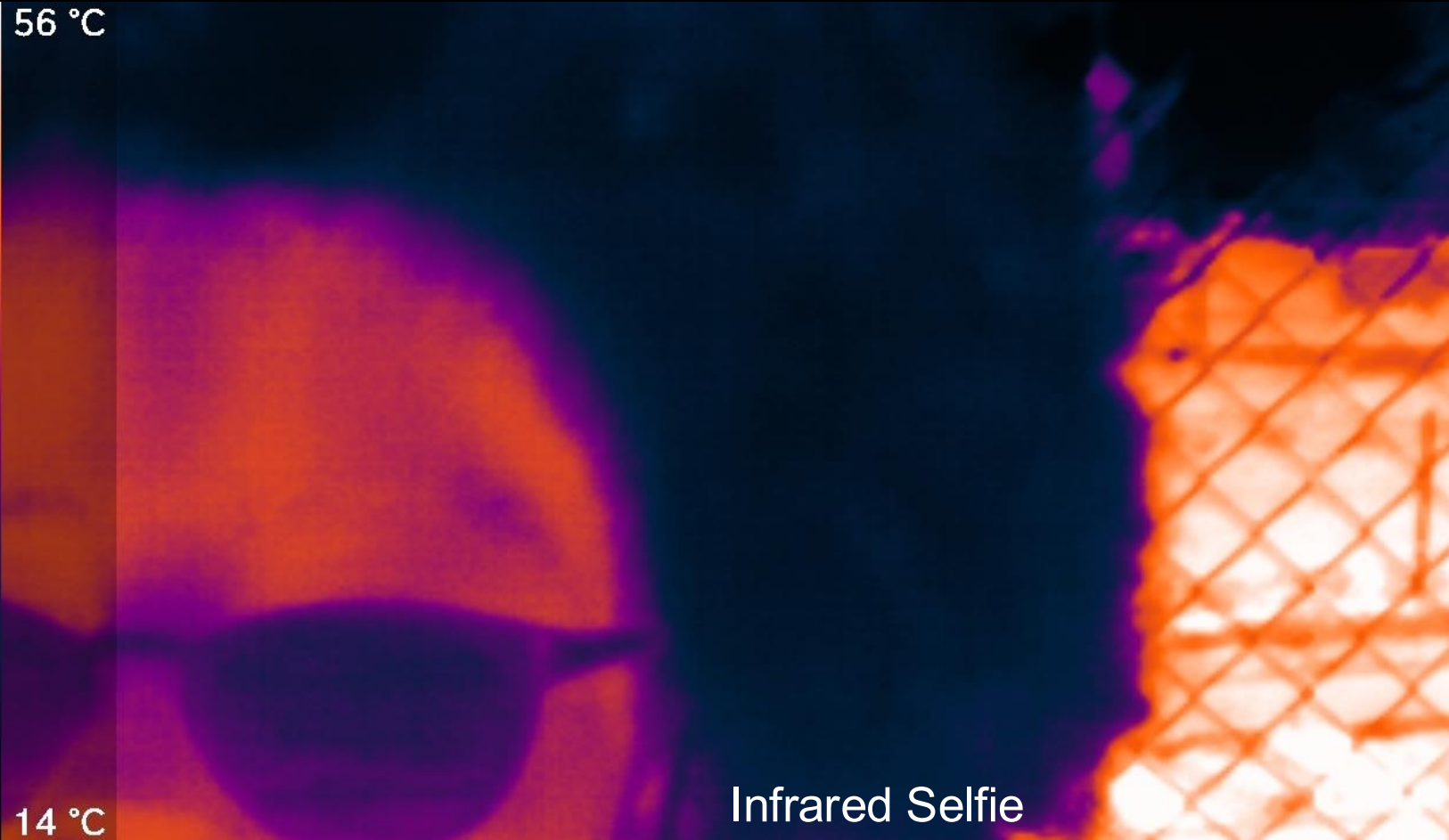
Filters

Astronomical filters limit light from the telescope to a specific bandpass.

The bandwidth of a filter is described by the distance between the 50% transmission values along the filter curve. This is called the full width at half maximum (FWHM).

Filter choice is essential to a successful observing program.

Wavelength/Filter Choice Matters



Infrared light tells us a lot about heat. Glass usually does not transmit Infrared light, so you can't see my eyeballs.

Example: Jupiter in Different Filters



Infrared
4700 nm
Gemini / NIRI

IR (Heat)



Visible
395/502/631 nm
Hubble Space Telescope / WFC3

Optical (Your Eyes)



Ultraviolet
343/275/225 nm
Hubble Space Telescope / WFC3

Ultraviolet

We take pictures of planets in different wavelengths to understand their atmospheres. We want to answer questions like: Is this planet like Earth?, Venus?, what molecules are in the atmosphere?